

TIPPING POINTS, THRESHOLDS AND THE KEYSTONE ROLE OF PHYSIOLOGY IN MARINE CLIMATE CHANGE RESEARCH

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Abstract

The ongoing and future effects of global climate change on natural and human-managed ecosystems have led to a renewed interest in the concept of ecological thresholds or tipping points. While generalizations such as poleward range shifts serve as a useful heuristic framework to understand the overall ecological impacts of climate change, sophisticated approaches to management require spatially and temporally explicit predictions that move beyond these oversimplified models. Most approaches to studying ecological thresholds in marine ecosystems tend to focus on populations, or on non-linearities in physical drivers. Here we argue that many of the observed thresholds observed at community and ecosystem levels can potentially be explained as the product of non-linearities that occur at three scales: (a) the mechanisms by which individual organisms interact with their ambient habitat, (b) the non-linear relationship between organismal physiological performance and

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variables such as body temperature and (c) the indirect effects of physiological stress on species interactions such as competition and predation. We explore examples at each of these scales in detail and explain why a failure to consider these non-linearities – many of which can be counterintuitive – can lead to Type II errors (a failure to predict significant ecological responses to climate change). Specifically, we examine why ecological thresholds can occur well before concomitant thresholds in physical drivers are observed, i.e. how even small linear changes in the physical environment can lead to ecological tipping points. We advocate for an integrated framework that combines biophysical, ecological and physiological methods to generate hypotheses that can be tested using experimental manipulation as well as hindcasting and nowcasting of observed change, on a spatially and temporally explicit basis.



1. INTRODUCTION

Anthropogenic climate change is among the most critical threats facing the world's natural and human-managed ecosystems (Rockstrom *et al.*, 2009). Numerous studies have documented geographic shifts in species range boundaries (Beaumont and Hughes, 2002; Parmesan and Yohe, 2003; Zacherl *et al.*, 2003), alterations in phenology (Parmesan, 2006; Mitchell *et al.*, 2008) and episodes of mass mortality (Harley, 2008; Harley and Paine, 2009) related to climate change. While temperature is one of the more obvious drivers of these patterns (Tomanek, 2008; Pörtner, 2010; Somero, 2010), stressors such as ocean acidification (OA) (Fabry, 2008; Hoegh-Guldberg and Bruno, 2010) have also been shown to have significant ecological impacts. The biological effects of climate change in turn have enormous economic and societal implications (Climate Change Science Program, CCSP, 2008; Millennium Ecosystem Assessment, 2005). Subsequently, and as highlighted recently by Hoegh-Guldberg and Bruno (2010), there has been an increase in the number of peer-reviewed papers examining climate change and its consequences to the natural world and to human society (Fig. 3.1).

While generalizations such as poleward migrations of species range boundaries or upward shifts in altitudinal distribution in mountain environments have served as a useful heuristic framework for exploring the ecological impacts of climate change, recent evidence suggests that these generalizations may be violated in nature more often than has previously been appreciated. For example, Crimmins *et al.* (2011) found that despite increases in ambient air temperature, the distribution of 64 plant species shifted downward in elevation during the last *ca.* 75 years. This pattern was explainable because the negative physiological impacts of increased temperature were overridden by the positive impacts of increases in precipitation, which had also occurred during this time period. Similarly,

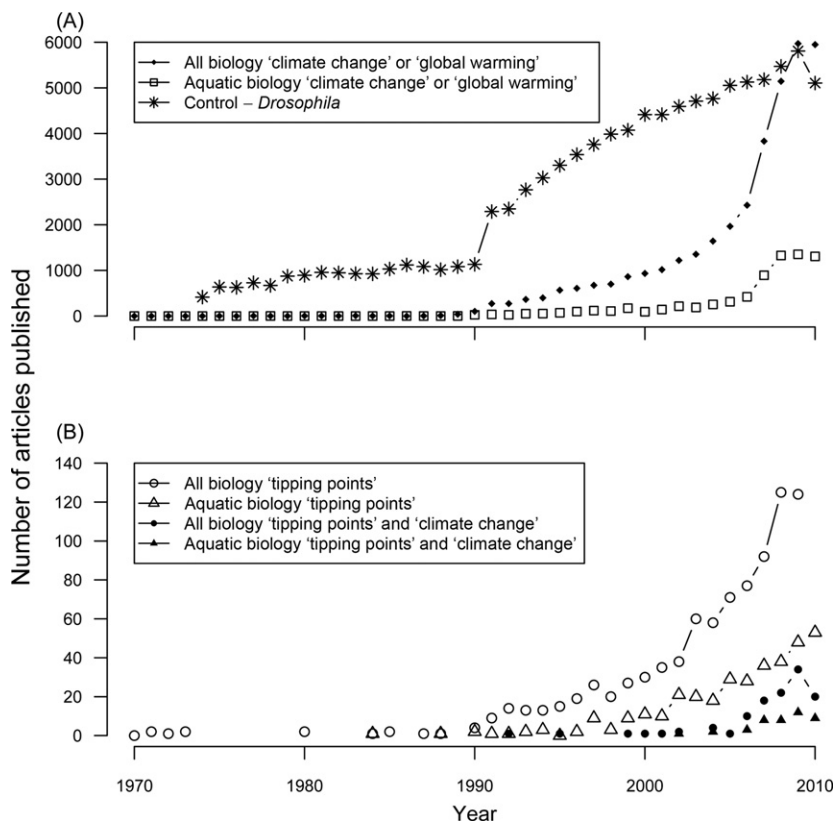


Figure 3.1 Number of peer-reviewed articles related to climate change and tipping points that have been indexed by the ISI Web http://apps.webofknowledge.com/WOS_GeneralSearch_input.do?SID=S2P9GLPhpF6fKNgfEp&product=WOS&search_mode=GeneralSearch of Science between 1970 and 2010. (A) Annual number of papers indexed with the topic 'climate change' or 'global warming', restricted to all fields related to biology as well as those restricted to aquatic biology. As a control for both the increase in journals as well as bias due to the database itself, the number of citations for the term '*Drosophila*' is also presented. Note that while there is a large overall increase in publications after 1990 (as indicated by the 'control'), the increase in the number of these publications appears to be slowing; in contrast, the increase in the rate of publication of climate-related papers began much later, and is increasing rapidly, as shown by [Hoegh-Guldberg and Bruno \(2010\)](#). (B) General biology and aquatic biology articles that have utilized the concepts 'tipping points', 'ecological thresholds' or 'stable states' independently, or in combination with 'climate change' or 'global warming' in papers restricted to the ecological and environmental literature.

numerous studies have now shown that geographic patterns of physiological stress do not always follow simple latitudinal gradients, but rather exhibit 'mosaic' patterns over geographic scales ([Helmuth et al., 2002, 2006](#); [Holtmeier and Broll, 2005](#); [Finke et al., 2007](#); [Place et al., 2008](#);

Mislan *et al.*, 2009; Pearson *et al.*, 2009), suggesting that our expectation of what to expect in coming decades may not always be poleward shifts in species range boundaries, but at least in some cases may be localized extinctions even well within range boundaries. Moreover, recent studies have documented geographic variability in physiological tolerance (Pearson *et al.*, 2009) and have experimentally shown evidence of local adaptation (Kuo and Sanford, 2009). All of these studies suggest that a detailed understanding of the mechanisms underlying the complex interaction between changes in the physical environment and organismal and ecological responses is vital if we are to predict future patterns of biodiversity, distribution and abundance, and that simple generalizations are not always effective as working null hypotheses.

Moreover, increasingly sophisticated adaptation planning by a wide array of decision makers demands quantitative predictions of ecological impacts of climate change, often at fine spatial and temporal resolutions, along with associated estimates of uncertainty. For example, the emplacement of protected areas (Hoffman, 2003), predictions of fishery, crop and livestock productivity (CCSP, 2008) and estimates of the spread of disease and invasive species (Chown and Gaston, 2008; Kearney *et al.*, 2008; US EPA, 2008) all demand quantitative, spatially and temporally explicit predictions of how climate change is likely to impact organisms and ecosystems (Ludwig *et al.*, 2001). Predictions that are based on simple generalizations are highly unlikely to be able to capture the spatial and temporal complexity of the real world in order to effectively plan and prepare for ongoing and future climate change impacts.

Furthermore, the public's perception of climate change, and their trust in the scientific community's understanding of climate change, is significantly affected by the frequency by which scientific predictions are borne out. Simple generalizations, while logically tractable, are not always the expected outcome, as shown by Crimmins *et al.* (2011). Nevertheless, when patterns contrary to sweeping generalizations are reported, they are often viewed by 'climate skeptics' as evidence of the falsehood of climate change or, worse, of the duplicity of scientists. In a recent opinion piece, Parmesan *et al.* (2011) stated that any attempts to attribute individual responses to climate change were 'ill advised,' suggesting that because such cause and effect linkages were too complex, the scientific community should instead focus on overarching trends. In stark contrast, however, data collected by communication experts have shown that when scientists make explicit predictions that can then be tested in a transparent manner, even at the risk of failure, this builds trust between the public and the scientific community (Goodwin and Dahlstrom, 2011). Thus, explicit, testable predictions made using mechanistic approaches (Helmuth *et al.*, 2005) not only provide useful information to decision makers, and potentially elucidate important principles by which weather and climate affect organisms

and ecosystems, but may also play an important role in discourse between scientists and the public.

While several recent authors have highlighted the importance and power of cross-disciplinary research when exploring the ecological impacts of climate change (Wiens and Graham, 2005; Stenseth, 2007; Chown and Gaston, 2008; Denny and Helmuth, 2009; Hofmann *et al.*, 2010), in many ways such collaborations have yet to be fully realized. To a large extent, environmental data are collected, archived and disseminated with little or no thought given to their potential biological applications (Helmuth *et al.*, 2010). Similarly, many physiological studies are conducted with only a limited ecological context, and vice versa (Denny and Helmuth, 2009). When such studies are used in the decision-making process, they also frequently fail to take the needs of end-users into consideration (Agrawala *et al.*, 2001).

Here we discuss some of the underlying reasons why an understanding of the detailed mechanisms by which weather and climate (both terrestrial and oceanic) affects organisms and ecosystems can fundamentally alter our predictions of the ecological impacts of climate change. We focus primarily on coastal marine invertebrates using the concept of ‘tipping points’ (ecological thresholds) to explore the importance of these details. We advocate for an interdisciplinary, mechanistic approach, explicitly embedded within a collaborative framework that combines assessments of physiological performance, organismal biology and population and community ecology performed at different scales. Specifically, we argue that many of the observed instances of tipping points are explainable given non-linearities in how environmental signals are translated into physiological responses, and subsequently how those physiological responses drive species interactions and population dynamics that affect ecosystem-level patterns. We divide these non-linearities into three broad categories: (a) the translation of environmental parameters (‘habitat’, as in Kearney, 2006) into niche-level processes such as body temperature; (b) the physiological consequences of these niche-level processes and (c) the indirect, ecological effects of physiological stress on species interactions.

The goal of this chapter is not to provide an exhaustive review of studies of ecological thresholds in marine ecosystems: a recent special issue of *Marine Ecology Progress Series* (Osman *et al.*, 2010) provides an excellent overview of the current state of the field. Nor is it our intention to imply that studies or approaches that focus on populations or ecosystems rather than on individual organisms and physiological responses are flawed. For example, catastrophic physical disturbance such as damage from hurricanes obviously can play a large role in driving phase shifts and may have no connection to physiological performance (although, physiological performance may contribute to recovery from such events, *sensu* Highsmith *et al.*, 1980). However, current discussions of the concept of ecological thresholds/tipping points seldom incorporate the physiological performance of the constituent organisms, despite an increasing number of studies focused on ecological thresholds in

the context of global climate change (Fig. 3.1B). For example, the CCSP of the United States, recently released the Synthesis and Assessment Product 4.2 (SAP 4.2), *Thresholds of Climate Change in Ecosystems*, a comprehensive report that specifically elaborates on our current knowledge of ecological thresholds at the ecosystem level and provides guidelines for resource managers who are forced to contend with the uncertain scenarios presented by global climate change. The document lists areas where further research is needed to fill the many gaps in our current understanding of the causes and consequences of ecological thresholds (CCSP, 2009), and advocates for an integrative approach as a means for dealing with cross-scale interaction processes. Notably, however, because the report primarily focuses on large-scale ecosystem responses, it generally fails to recognize the importance of examining the impacts of weather and climate at organismal scales (Somero, 2010). Our goal is therefore to demonstrate how much can potentially be learned through an integrated approach that includes an understanding of the mechanisms underlying thresholds, including biophysical interactions between organisms and their environment, and physiological consequences of climate change at organismal scales (Sanford, 2002a; Pörtner *et al.*, 2006).

1.1. Non-Linearities, tipping points and concepts of scale

A system is said to be non-linear when its inputs and outputs are disproportionate to one another (Hilbert, 2002). Within an ecosystem context, the term input thus refers to any relevant abiotic variable, e.g. precipitation or air temperature, or a biotic component such as a keystone predator (Menge *et al.*, 1994), and the output is any physiological or ecological process such as an organism's phenology, reproductive output or ecological interactions (Sanford, 2002a; Pincebourde *et al.*, 2008). The CCSP SAP 4.2 report defines an ecological threshold (tipping point) as 'the point where there is an abrupt change in an ecosystem quality, property, or phenomenon or where small changes in an environmental driver produce large, persistent responses in an ecosystem, which is not likely to return to the previous more stable state' (CCSP, 2009).

Studies of ecological thresholds are extremely valuable for planning conservation strategies, as they can shed light on an ecosystems' sensitivity to environmental change (Littler and Littler, 2007; Briske *et al.*, 2010). Studies performed at the community and ecosystem levels of organization have also detected the presence of tipping points that set the boundary between different ecological states (Scheffer *et al.*, 2001), although other authors have suggested that the simple dichotomy between alternative stable states, while intellectually appealing, may be an oversimplification of a much more complicated process (Dudgeon *et al.*, 2010). Long-term observations of community-level dynamics have allowed for both empirical and theoretical descriptions of major phase shifts and/or alternative

stable states (Hare and Mantua, 2000; Casini *et al.*, 2009; Dudgeon *et al.*, 2010) in marine systems such as coral reefs (Idjadi *et al.*, 2006) and rocky intertidal communities (Petraitis *et al.*, 2009). Characterizing these higher-level changes has proven useful for ecologists and wildlife managers, since they provide an overall understanding of the main environmental variables shaping natural systems.

So far, most studies of threshold effects in marine systems have measured and modelled processes at community and ecosystem scales. Empirical studies that only concentrate on higher-level dynamics can seldom tease apart the underlying factors driving systems to change, which often restricts conclusions to pure documentations of the observed patterns. Of course, a focus on community- and ecosystem-level processes does not mean that authors do not recognize the importance of underlying biological processes at the scale of the organism (Hewitt and Thrush, 2010). For example, Norkko *et al.* (2010) manipulated disturbance via hypoxia at scales ranging from 1 to 16 m² and monitored recovery by infaunal and epifaunal organisms living in soft substrate communities. Their results highlighted the importance of considering life-history characteristics (epifaunal/infaunal) and mobility (dispersal) of the organisms in driving the resilience and recovery of the benthic community. There are also excellent examples of the power of considering physiological performance in the context of ecological thresholds (Littler and Littler, 2007; Hofmann *et al.*, 2010). Most notably, recent studies of OA squarely place an emphasis on measuring the effects of decreases in pH on growth and survival of ecologically key species when attempting to understand and predict ecological consequences at large scales (Fabry, 2008; Hofmann *et al.*, 2010). For example, McNeil and Matear (2008) measured and modelled levels of carbonate (CO_3^{2-}) and pH in the southern ocean, and then projected the impact on rates of calcification by key planktonic species. Their results suggested that the pteropod *Limacina helicina* Phipps, 1774, was likely to be severely impacted during larval development. This species is ecologically important, comprising up to ~65% of the total zooplankton in the Ross Sea, and the thecosome shells of this and other pteropod species are thought to be a major contributor to the carbonate flux of the deep ocean south of the Polar Front (Hunt *et al.*, 2008). Understanding the physiological responses of calcifying organisms such as pteropods, corals (Hoegh-Guldberg *et al.*, 2007) and coccolithophores (Fabry, 2008) to changes in ocean pH thus clearly has significant ecological consequences, and therefore physiological performance is likely to have a direct impact on the probability of phase transitions/ecological thresholds occurring.

Many recent studies have further emphasized the importance of considering physiological performance in determining local and biogeographic patterns of distribution (Somero, 2005; Helmuth *et al.*, 2006; Pörtner *et al.*, 2006; Gedan and Bertness, 2009), an approach often termed

‘macrophysiology’ (Chown *et al.*, 2004; Chown and Gaston, 2008). For example, Wetthey and Woodin (2008) compared long-term records of sea surface temperature against known physiological thresholds of the barnacle *Semibalanus balanoides* and showed that observed range shifts were consistent with winter temperatures known to cause reproductive failure in this species. However, studies that span scales from molecular to biogeographical remain rare (Pörtner *et al.*, 2006; Denny and Helmuth, 2009; Pearson *et al.*, 2009; Hofmann *et al.*, 2010). Moreover, a number of studies have shown that organisms may be living close to their physiological tolerances even well within their range limits (Sagarin and Somero, 2006; Place *et al.*, 2008; Beukema *et al.*, 2009), and have warned that, conversely, physiological stress is not always the limiting factor at species range edges (Davis *et al.*, 1998a,b). Thus, while understanding the relationship between the physiological performance of key species and ecological thresholds may not always be simple (Hutchins *et al.*, 2007; Crain *et al.*, 2008), a failure to consider these effects can potentially lead to Type II errors, i.e. we may be surprised by sudden phase shifts due to non-linearities that ultimately originate at (sub)organismal scales. Understanding when such events are likely to occur, and the mechanisms that lead to their occurrence, is therefore critical. Perhaps even more importantly, as recently discussed by Mumby *et al.* (2011), tipping points may be preceded by significant alterations in ecosystem function that, while not meeting the definition of a ‘threshold’, nevertheless may have catastrophic ecological, economic and societal implications. For example, declines in ecosystem services may occur well before threshold events are observed. As a result, Mumby *et al.* (2011) argue that while threshold events are important, we must not lose focus on the importance of predicting declines in other metrics of ecosystem performance. In this review we explore how small, often linear changes in physical drivers may potentially lead to large, non-linear responses in ecological systems. As a result, ecological tipping points may theoretically occur long before any comparable changes in the physical environment are observed. However, such methods may also be applied to the agenda set forth by Mumby *et al.* (2011), in that they provide a mechanistic framework that can be used to predict potential ‘trouble areas’ not just in terms of threshold events but also in terms of declining ecosystem function.



2. WEATHER, CLIMATE AND CLIMATE CHANGE FROM THE VIEWPOINT OF A NON-HUMAN ORGANISM

Climate change is a global phenomenon, but to an organism the ‘world’ can be exceedingly small. Consider for example an intertidal barnacle. As a cyprid floating in the water column, only the immediate

conditions of pH, temperature and food surrounding the animal affect its physiology. To that larval animal, it does not matter if it is entrained in a gyre or in the nearshore swash zone *per se*, but rather what its location within either of those larger-scale phenomena means to its immediate physical and biological environment. As the animal moves onshore, it encounters levels of turbulence, temperature, pH and nutrients different from those in the immediate nearshore environment (Pineda and Lopez, 2002; Pfister *et al.*, 2007; Wootton *et al.*, 2008). Importantly, those conditions likely could not have been predicted given measurements made just offshore (Pfister *et al.*, 2007). Eventually, as the larva reaches the intertidal zone where it settles, metamorphoses and grows into an adult barnacle, it experiences not only the conditions of the subtidal environment, but also those of the terrestrial environment during low tide, conditions which can vary markedly with intertidal zonation height (Wethey, 1983). Perhaps not surprisingly, all point to the fact that the physical environment for these animals, like that for many others, is highly spatially and temporally heterogeneous (Denny *et al.*, 2004, 2011).

Nevertheless, in many cases, measurements conducted at moderate to large spatial scales, e.g. by satellite and buoy, appear to provide considerable insight into large-scale ecological processes (Schoch *et al.*, 2006; Blanchette *et al.*, 2008; Gouhier *et al.*, 2010). Similar concordance appears over large temporal scales, and when looking at the ecological influences of climate indices such as the El Niño Southern Oscillation (ENSO), the North Atlantic Oscillation (NAO) and the Pacific Decadal Oscillation (PDO; Stenseth *et al.*, 2002, 2003; Forchhammer and Post, 2004). However, as pointed out by Hallet *et al.* (2004), the seemingly superior ability of these large-scale indices to predict biological responses than higher frequency weather data may lie in a failure to take mechanism into account. Using detailed data from a population of Soay sheep, Hallet *et al.* (2004) showed that high rainfall, high winds or low temperatures could all contribute to the mortality of sheep, either immediately or with a lag. In other words, the association between each of these variables and the timing of mortality varied significantly between years, so that overall there appeared to be no pattern. Without an understanding of the underlying physiological mechanistic drivers, simple correlations between any single variable such as rainfall and mortality failed to uncover any relationship, giving the false impression that climatic indices were a better predictor of ecological response than were weather variables. However, when variables such as rainfall and air temperature were used in an integrated context that considered not only their direct physiological effects but also their indirect effects on food, etc., a highly significant relationship emerged (Hallet *et al.*, 2004) that had greater explanatory power than did climatic indices. Such may be the case for many environmental factors that are often dismissed as irrelevant due to their apparent lack of

concordance with biological processes: by assuming a direct relationship between variables such as solar radiation, air and water temperature, or food availability and population responses, we fail to consider that these variables interact in non-linear ways, and are filtered through the functional traits of the organisms that they are affecting (Kearney, 2006; Kearney *et al.*, 2010).

A case in point is how weather variables are translated into physiologically and ecologically relevant terms such as body temperature (Kearney, 2006). While we may be interested in climate change, it is weather (as affected by climate) that drives physiological responses. Global climate change encompasses change in numerous 'environmental signals' (Helmuth, 2009) including ocean pH, sea level, salinity and temperature. Importantly, however, the only environmental signals that matter to an organism directly are those that the organism experiences, i.e. those at the level of the niche (Kearney, 2006). The physiological niche of an organism is driven by the interaction of an organism's morphology, size and behaviour with its local microhabitat, and is therefore often very different from measurements of large-scale, habitat-level parameters such as air or water temperature (Marshall *et al.*, 2010; Helmuth *et al.*, 2006, 2011). Thus, two species inhabiting the same microhabitat may experience radically different physiological drivers such as body temperature. For example, as has been shown for many species in both terrestrial and intertidal environments, the flux of heat is driven by the interaction of multiple environmental factors, including solar radiation, wind speed, air temperature and relative humidity (Porter and Gates, 1969; Bell, 1995; Marshall *et al.*, 2010). Moreover, the characteristics of the organism – mass, colour, surface wetness, etc. – significantly alter heat flux so that two organisms exposed to identical environmental parameters can experience markedly different body temperatures (Broitman *et al.*, 2009). In some cases, the difference between body temperature and environmental temperature can be fairly minor, e.g. when animals continually live in the shade or in deep water environments.

In other cases, the difference in temperature between an ectothermic organism and its surroundings is quite remarkable. The body temperatures of ectothermic animals in the sun are generally significantly hotter than the temperature of the surrounding air (Marshall *et al.*, 2010). Only in cases where animals lose heat through the evaporation of water, or through infrared radiation at night, is body temperature likely to be significantly lower than that of air or surface temperature (Bell, 1995; Helmuth, 2002). These observations are significant, because they suggest that for many organisms (e.g. those that are unable to evaporatively cool or for which desiccation is a limitation), air temperature is likely to set the *lower* limit to body temperature during the day, and solar radiation then increases body temperature above that minimum. Critically, this also means that convective heat transfer serves to bring the temperature of the

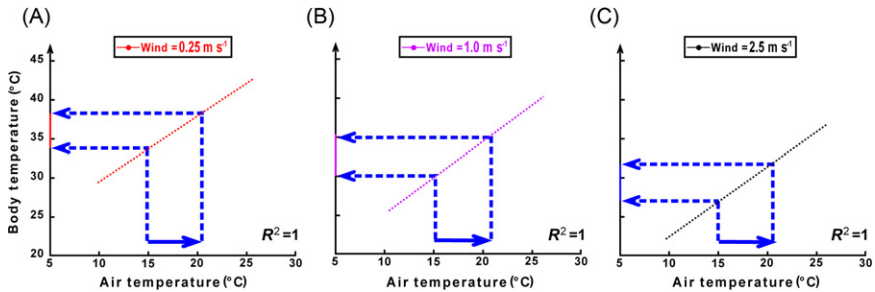


Figure 3.2 Translation of habitat-level parameters such as air temperature to niche-level parameters (which drive physiology) such as body temperature. The figure shows a steady-state heat budget for a generic ectotherm under identical conditions of solar radiation and cloud cover, over a range of air temperatures, and with wind speeds of (A) 0.25 m s^{-1} , (B) 1.0 m s^{-1} and (C) 2.5 m s^{-1} . In this simplified example, body temperature increases linearly with increasing air temperature. Notably, the y intercept varies with wind speed, so that a 5°C increase in air temperature results in shifts between very different magnitudes of body temperature.

animal closer to that of the surrounding air, i.e. cooler, even in the absence of evaporation. For example, recent modelling suggests that predicted increases in the mean wind field along the west coast of the United States may in some cases counteract the effect of increases in ambient air temperature on animal body temperature, at least for animals with dry surfaces (Helmuth *et al.*, 2011). In contrast, for animals with wet surfaces (seastars, seaweeds), the forecasted increase in mean wind may have a greater physiological impact through increased desiccation stress (Bell, 1995). Importantly, this does not mean that changes in parameters such as air temperature are unimportant; if all other environmental factors remain unchanged, increases in air temperature will lead to increases in body temperature (Fig. 3.2). However, in some cases the importance of variability or long-term change in air temperature can be overridden by other factors such as wind speed, wave splash (Helmuth *et al.*, 2011) or the timing of when low tide occurs (Mislan *et al.*, 2009).

Moreover, these results indicate that increases in body temperature cannot be based on changes in any one environmental parameter. Figure 3.2 shows the results of a simple heat budget model for a generic ectotherm, in which all parameters are identical except for wind speed. In the three simplified scenarios shown, body temperature increases linearly with air temperature, i.e. the coefficient of determination is 1.0. However, the y intercept varies markedly depending on which value of wind speed is used. In the first scenario (wind speed = 0.25 m s^{-1}), an increase in air temperature from 15°C to 20°C leads to an increase in body temperature from 33°C to 38°C ; in the second scenario, the same change in air

temperature, but with a wind speed of 1.00 m s^{-1} , leads to a change in body temperature from 30°C to 35°C ; in the third scenario, with a wind speed of 2.50 m s^{-1} , the same change in air temperature leads to a change in body temperature from 26°C to 31°C . Clearly, simply measuring the temperature of the habitat (air), or even the change in the air temperature, is not sufficient to assess how the changing environment will impact the organism.

In the steady-state scenario shown in Fig. 3.2, all parameters except wind speed are held constant, and body temperature increases linearly with air temperature. Under natural field conditions it is unclear how often this holds true, but in the few cases where explicit comparisons have been made it appears that the relationship between air and body temperature can be extremely poor. Marshall *et al.* (2010) compared maximum air temperature at low tide to the body temperature of intertidal snails and found that there was no correlation between the two temperatures; at times the temperature of the animal could be 22°C above that of the air. Helmuth *et al.* (2011) compared maximum air temperature at low tide to the temperature of biomimetic loggers designed to mimic the temperature of intertidal mussels and reported a coefficient of determination of only 0.14, with differences of up to 19°C between air and animal temperature. They also showed that mussel (logger) temperatures were frequently high even on days when air temperatures were low and vice versa. Broitman *et al.* (2009) compared the temperatures of predators (*Pisaster ochraceus* Brandt, 1835) to those of their prey (*Mytilus californianus* Conrad, 1837) under identical microclimatic conditions at four sites, two on each end of Santa Cruz Island, CA. They found that on the west side of the island, the temperatures of predators and prey were very similar, but on the east side of the island, the body temperatures of the predator and prey were very different from one another, even though in all cases both species were exposed to identical weather conditions at each site. Again, these results point to the interaction between multiple weather variables, and between weather and the functional traits of the organism (Kearney *et al.*, 2010) in driving the body temperature of ectotherms and demonstrate the highly non-linear nature of the relationship between 'habitat' and 'niche' (Kearney, 2006).

While arguments regarding heat flux apply most directly to terrestrial organisms and intertidal organisms exposed to air at low tide, studies have shown that in shallow water, solar radiation can raise the temperature of coral tissue by several degrees when rates of convection (i.e. water flow) are sufficiently low (Fabricius, 2006; Jimenez *et al.*, 2008). Analogous arguments can also be made for the exchange of gas and nutrients in subtidal environments, where the flux of these substances is driven not only by concentration gradients but also by fluid flow (Lesser *et al.*, 1994). As elegantly discussed by Patterson (1992), the characteristics of fluid flow

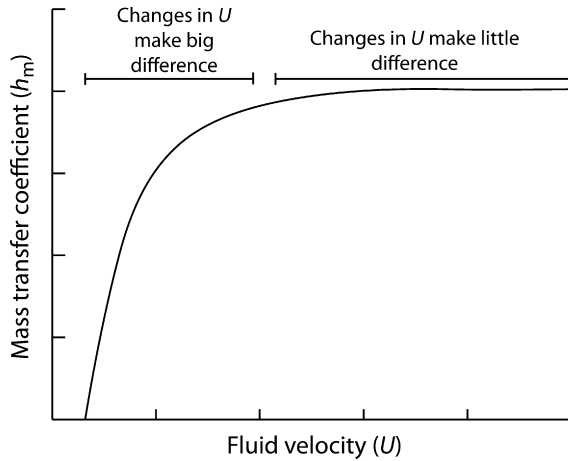


Figure 3.3 Typical mass transfer coefficient relationship as a function of flow speed, in which small increases initially make a large difference, but past a threshold have little effect (heat transfer coefficients show a very similar relationship).

around the respiratory and feeding structures of subtidal organisms determine rates of exchange of oxygen, bicarbonate and nutrients through their effects on the diffusion boundary layer (Shashar *et al.*, 1996). A generic equation to describe the exchange of any mass item thus includes a mass transfer coefficient (h_m), an empirically derived parameter that describes the interaction of an organism with the surrounding flow:

$$dm/dt = h_m A (C_o - C_i) \quad (3.1)$$

where dm/dt is the rate of mass flux; h_m is the mass transfer coefficient, A is the area over which exchange occurs, and C_o and C_i are the concentrations of the mass item of interest (e.g. O_2) outside and inside of the organism, respectively. The mass transfer coefficient changes non-linearly with increasing flow (Fig. 3.3), and is lower for ‘streamlined’ animals than for animals with a bluff body. (The equation for convective heat exchange is functionally identical, except that a heat transfer coefficient, h_c , is used to describe the effect of morphology on the rate of heat exchange, which is affected by a temperature gradient rather than a concentration gradient.)

Numerous studies have documented the important role of mass flux in driving the physiological ecology of benthic organisms. Both laboratory (Nakamura *et al.*, 2003) and field studies (Nakamura and van Woesik, 2001; Finelli *et al.*, 2006) have shown that increased mass flux reduces the rate of coral bleaching through the removal of excess oxygen, which reduces oxidative stress (Lesser, 1996, 1997, 2004). As with heat exchange in the terrestrial environment, the morphology of an organism can significantly affect fluid flow, so that two organisms exposed to identical flows,

and/or identical concentrations of gases, can experience very different rates of passive gas uptake. [Lenihan *et al.* \(2008\)](#) showed an effect of reef structure (height and morphology) on rates of bleaching in Moorea, French Polynesia. At a much smaller scale, [Finelli *et al.* \(2007\)](#) found that intracolony variability in coral bleaching could be explained by differences in flow, which in turn was affected by coral morphology. [Thomas and Atkinson \(1997\)](#) showed that rates of flow and surface roughness controlled rates of ammonium uptake by corals. These results demonstrate why simply measuring the concentration of oxygen, bicarbonate or ammonium is not sufficient to estimate rates of exchange; two organisms exposed to identical 'habitats' (gas or nutrient concentrations) will experience markedly different rates of uptake depending on how their functional traits/morphology interact with local flow.

Directly analogous studies have shown the non-linear relationship between water flow and the risk of dislodgement by sessile organisms, especially those in wave-swept environments (Denny and Gaylord, 2010). The force of drag acting upon an organism varies as the square of water velocity, so that small changes in flow can result in large changes in the force of drag acting on an organism:

$$\text{Drag} = \frac{1}{2}\rho AC_d U^2 \quad (3.2)$$

where A is the area upon which the fluid acts, ρ is fluid density, C_d is the drag coefficient, and U is fluid velocity. As above, the C_d is a function of organism morphology, and a bluff body has a much higher C_d than does a streamlined organism. Similarly, the force of lift also scales with the square of U (Denny and Gaylord, 2010). Numerous studies have examined the interactions between the fluid environment and the risk of dislodgement of sessile organisms, noting that one prediction of climate change in many regions is an increase in wave height ([Boller and Carrington, 2007](#); [Carrington *et al.*, 2009](#)). While some studies have suggested that the relationship between increasing wave height and onshore wave velocity may be more complex than expected due to the tendency of larger waves to break farther offshore ([Helmuth and Denny, 2003](#)) and because of the overwhelming importance of small-scale topography in affecting local flows ([Denny *et al.*, 2004](#)), these results nevertheless point to the potentially important, highly non-linear relationship between small increases in wave height, water flow and the risk of dislodgement.

Thus, almost never are interactions between organisms and their physical environment, or interactions between the physical parameters acting on organisms, linear in their impacts. For example, heat and mass transfer coefficients, which describe the interactions between the morphology of an organism (or colony) and the surrounding fluid in driving heat or mass (e.g. gases and nutrients), are usually a nearly asymptotic function, in which small changes in fluid velocity initially lead to a large change in

exchange of mass or heat. After a point, however, further increases in flow lead to very little change in heat or mass flux (Fig. 3.3). Small increases in wind speed initially have a large effect of cooling through convection, but once wind serves to bring the temperature of an organism close to the temperature of the surrounding air, very little change occurs with further increases in wind speed. Likewise, small increases in water flow can initially lead to large increases in gas or nutrient flux, but past some threshold make little difference. Conversely, when conditions of wind or flow decrease, initial changes may result in little change when the range of conditions corresponds to the 'plateau' region of the curve, but below some threshold, rapid changes may ensue with even subtle drops in flow. Note that these relationships starkly contrast with the relationship between water flow and drag, which increases exponentially. In essence, therefore, these non-linearities thus create tipping points as environmental parameters at the habitat level are translated into changes at the niche level (Kearney, 2006): linear changes in parameters such as flow speed, wave height or air temperature can lead to non-linear changes in physiologically relevant parameters such as body temperature or gas flux, and thus to the likelihood of reproductive failure and mortality of key species.



3. PHYSIOLOGICAL RESPONSE CURVES

Although changes in niche-level responses such as body temperature and oxygen exchange are the proximal drivers of physiological response, ultimately cellular- and subcellular-level reactions will determine the consequences of those environmental changes (Somero, 2010). For example, Carrington (2002) has shown that the attachment strength of mussels (*Mytilus edulis* Linnaeus, 1758) varies seasonally, and is twofold higher in winter than in summer. However, the match between wave force and attachment strength is not perfect, and during hurricane season mussels are only weakly attached. Their results suggest a potential energetic trade-off between the production of byssal threads and energy devoted to reproduction (Carrington, 2002), and emphasize the critical importance of measuring not only environmental variables but also the vulnerability of organisms to their physical environment (Helmuth *et al.*, 2005).

One of the best ways to describe an organism's response to changing environmental conditions is through the use of physiological performance curves. Physiological performance curves have long been used to define the complex relationships between organism responses related to fitness, such as growth, reproduction and survival and factors such as body temperature (although, notably, many studies mistakenly have used habitat

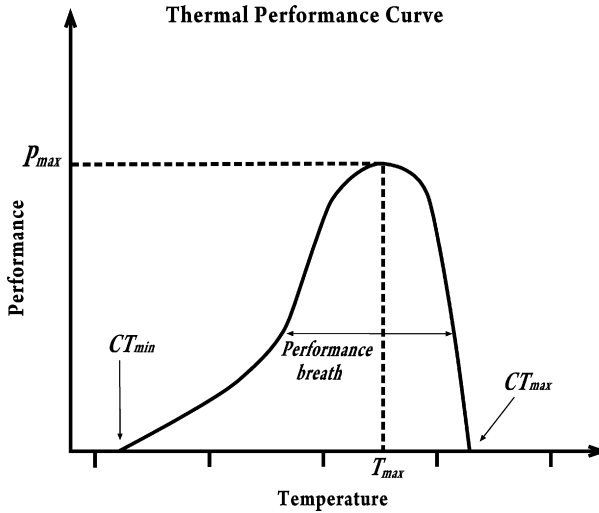


Figure 3.4 Standard physiological performance curve, using some aspect of performance related to fitness as a function of body temperature. P_{max} is maximum (optimum) performance, and T_{max} is the body temperature at which it occurs. CT_{min} and CT_{max} are the critical minimum and maximum temperatures, respectively. Performance breadth is defined based on a fixed percentage of P_{max} (here shown as a fairly small percentage for emphasis).

temperature as the independent axis, incorrectly assuming that it is equivalent to body temperature). Performance curves describe both an organism's physiological limits to survival and the conditions under which that organism can survive and reproduce (Huey and Stevenson, 1979; Angilletta *et al.*, 2002, 2003). Importantly, these curves are almost always 'left skewed' in that fairly large increases in body temperature (or other factor) above some lower threshold generally lead to only modest changes in performance (Fig. 3.4) until a maximal level of performance (P_{max}) is reached, at body temperature T_{max} (or T_{opt} ; Dewitt and Friedman, 1979; Angilletta, 2009). Above that optimum, however, performance declines rapidly with increasing temperature. Thus, at temperatures above T_{max} , small changes in body temperature can have significant impacts on survival and reproduction.

Returning to the scenario presented in Fig. 3.2, when a biophysical approach using a heat budget model is combined with a physiological performance curve, the importance of considering the non-linearities involved in how habitat-level parameters are translated into physiological responses becomes apparent. Under conditions of wind speed = 2.5 m s^{-1} , a 5°C increase in air temperature from 15°C to 20°C leads to an increase in body temperature from 26°C to 31°C (Fig. 3.5A). When this change in body (not air) temperature is translated to a physiological (thermal) performance

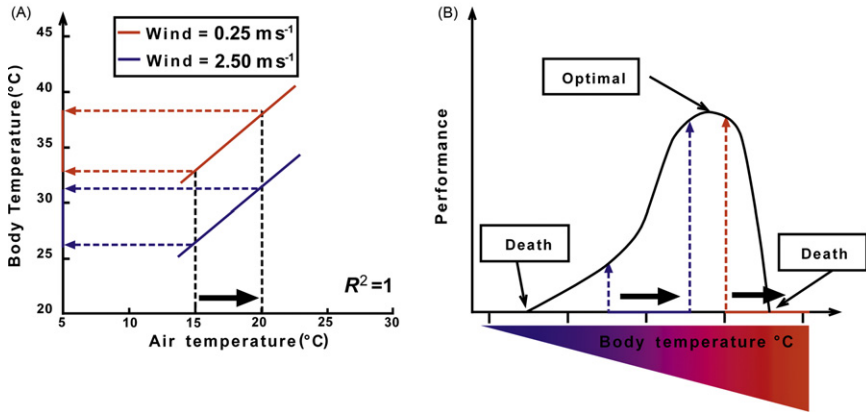


Figure 3.5 Importance of considering multiple environmental variables when predicting shifts in performance. (A) Influence of wind speed on how air temperature is translated to organism's body temperature, as illustrated in [figure 3.2](#). (B) Effect of body temperature change on the organism's performance. Under conditions of wind speed of 2.5 m s⁻¹, an increase in air temperature of 5°C (blue) leads to an increase in performance. Under conditions of wind speed of 0.25 m s⁻¹, an increase in air temperature of 5°C (red) leads to death.

curve, this leads to an increase in physiological performance, as the temperature of this hypothetical animal is brought closer to $T_{\max}$ ([Fig. 3.5B](#)). In contrast, an increase in air temperature from 15°C to 20°C but coupled with a wind speed of 0.25 m s⁻¹ leads to an increase in animal body temperature from 33°C to 38°C — enough to shift the animal's temperature from a point near its optimum to its lethal limit ([Fig. 3.5](#), in red).

3.1. Thermal physiology of marine organisms

Describing an organism's physiological performance curve under a range of physical conditions is considered a fairly straightforward analysis that can provide valuable information on how individual organisms respond to their environment, the energetic trade-offs that emerge from specific responses ([Angilletta et al., 2003](#)), and the evolutionary consequences of such responses ([Kingsolver et al., 2004](#)). However, while this approach has been used with great success by terrestrial ecologists, marine ecologists have rarely described a species' performance throughout its entire thermal range, especially for invertebrates. Although some excellent examples of marine species performance curves can be found in the literature ([Table 3.1](#)), marine invertebrate physiologists have primarily focused either on identifying thermal limits ($CT_{\max}$ and $CT_{\min}$), or on contrasting the effects of a few habitat temperatures covering only portions of a species' whole thermal range. Here we follow [Angilletta's \(2009, p. 36\)](#)

Table 3.1 Physiological performance traits of marine animals from a review of the literature

Phylum	Species	Performance trait	$P_{\max}$	$T_{\max}$	$CT_{\min}$	$CT_{\max}$	Citation
Mollusca	<i>Nucella lamellosa</i> adult	Crawling rate	40 cm h ⁻¹	20	0	30	1
Mollusca	<i>Nucella lamellosa</i> juvenile	Crawling rate	30 cm h ⁻¹	5–20	0	30	1
Mollusca	<i>Nucella ostrina</i>	Crawling rate	23 cm h ⁻¹	5–10	0	25	1
Arthropoda	<i>Mytilus</i> sp.	Speed of cilia	334 mm s ⁻¹	32.5	<0	40	2
Arthropoda	<i>Semibalanus balanoides</i>	Cirral activity	0.63 beats s ⁻¹	21	<2.3	31	3
Arthropoda	<i>Semibalanus balanoides</i>	Cirral activity	0.56 beats s ⁻¹	18.4	<1.8	31.5	3
Arthropoda	<i>Chthamalus stellatus</i>	Cirral activity	1 beats s ⁻¹	30	4.6	37.5	3
Arthropoda	<i>Balanus perforatus</i>	Cirral activity	0.9 beats s ⁻¹	25.2	6	36	3
Arthropoda	<i>Balanus perforatus</i>	Cirral activity	0.94 beats s ⁻¹	30.3	6	35.2	3
Arthropoda	<i>Balanus crenatus</i>	Cirral activity	1 beats s ⁻¹	21.3	<4.3	25.5	3
Arthropoda	<i>Elminius modestus</i>	Cirral activity	2.2 beats s ⁻¹	24.2	2	33	3
Arthropoda	<i>Lepas anatifera</i>	Cirral activity	0.28 beats s ⁻¹	19.8	0.5	33	4
Arthropoda	<i>Balanus improvisus</i>	Cirral activity	0.11 beats s ⁻¹	30	-2	35.5	4
Arthropoda	<i>Balanus amphitrite</i>	Cirral activity	0.14 beats s ⁻¹	29.9	6	38.4	4
Arthropoda	<i>Balanus balanus</i>	Cirral activity	0.48 beats s ⁻¹	20.2	-2	30	4
Chordata	<i>Botryllus schlosseri</i>	Reproductive output	2 larvae colony ⁻¹ week ⁻¹	25	15	>25	5
Chordata	<i>Botryllus schlosseri</i>	Growth rate	28 zooids colony ⁻¹ 70 day ⁻¹	20	5	>25	5
Chordata	<i>Botrylloides violaceus</i>	Growth rate	20 zooids colony ⁻¹ 70 day ⁻¹	20	5	>25	5
Chordata	<i>Salmo trutta</i> juvenile	Growth rate	0.3 g day ⁻¹ (1 g fish)	16.8	1.24	24.74	6
Chordata	<i>Salmo trutta</i> juvenile	Feeding rate	1.25 attempts min ⁻¹	17.3	<2.6	>24	6
Chordata	<i>Oncorhynchus nerka</i> juvenile	Growth	25% wet weight day ⁻¹	15	<1	14	7
Chordata	<i>Oncorhynchus nerka</i> adult	Growth	15% wet weight day ⁻¹	15	<1	14	7
Chordata	<i>Oncorhynchus nerka</i>	Food intake	27% body dry weight	17	<5	24	7

Temperatures are in °C.

References:

(1) Bertness and Schneider (1976), (2) Gray (1923), (3) Southward (1955a,b), (4) Southward (1957), (5) Epelbaum *et al.* (2009), (6) Ojanguren *et al.* (2001) and (7) Brett (1971).

definition of performance as ‘any measure of the organism’s capacity to function, usually expressed as a rate or probability’. The decision of which specific trait(s) to evaluate is of great importance for the conclusions that one can draw from response curves. In essence, one should consider the species’ niche-dependent requirements and measure those traits that most closely contribute to fitness (i.e. lifetime reproductive success). For instance, foraging rates and/or success are considered proxies of an organism’s fitness, but the way these are estimated is entirely dependent upon the organism’s life history. For example, the adult seastar *Leptasterias polaris* Müller and Troschel, 1842, a major subtidal predator inhabiting soft sediments in the northern Gulf of St. Lawrence (Canada), relies on its ability to sense the odour of infaunal prey to guide its digging efforts and to forage efficiently (Thompson *et al.*, 2005). Using odour sensitivity as a performance trait for *L. polaris* is consistent with its ecological niche; this trait would reflect a potential for survival and reproduction under different environmental conditions (e.g. current speed and water temperature). In contrast, for predators that rely on other traits to forage efficiently such as vision or speed, measuring odour sensitivity as a performance trait would not tell us anything about the mechanisms that drive their ecological responses.

Recent studies with marine species of crabs, fish, bivalves and polychaetes have used the concept of oxygen and capacity-limited thermal tolerance (Pörtner *et al.*, 1999; Sommer and Pörtner, 1999; Frederich and Pörtner, 2000; Pörtner and Farrell, 2008; Kassahn *et al.*, 2009;) to explain the mechanism that regulates both an organism’s thermotolerance windows (CTs) and thermal optima ($T_{\max}$). The theory states that detrimental temperatures bring about insufficient oxygen supply and transport to tissues, which coupled with high baseline oxygen demand at elevated temperatures likely shapes the typical ‘left-skewed’ thermal performance curve (Fig. 3.4; Angilletta, 2009).

Traditional life-history traits used to describe physiological performance curves include lifetime reproduction (e.g. fecundity and reproductive output), growth (e.g. change in size or body mass), feeding/assimilation (e.g. feeding rate and chemosensory ability), development rate and locomotion (e.g. speed and distance covered). Some investigators have also included survival probability – or any proxy for it such as righting response or burrowing capacity (Angilletta, 2009); we believe such studies are necessary for identifying temperature critical limits, but because they do not provide quantifiable information of an organism’s potential reproductive contribution (other than zero or ‘maybe’), we opt not to consider them as strict fitness-related performance traits.

From the traditionally measured traits listed above, lifetime reproduction is considered the most closely related to fitness; unfortunately quantifying the lifetime contribution of an individual through

reproduction is not an easy task (Angilletta, 2009), especially for broadcasting species with pelagic life-cycle stages, which are common in coastal marine ecosystems (Thorson, 1950). An example of a reproductive performance curve is given by Epelbaum *et al.* (2009) (Table 3.1), who tested the interactive effects of water temperature and salinity on two invasive species of colonial ascidians now present throughout much of the northeast Pacific, *Botryllus schlosseri* Pallas, 1766, and *Botryllus violaceus* Oka, 1927 (Lambert, 2005). The study included virtually the entire thermotolerance window, of the species (Brunetti *et al.*, 1980), and captured the conditions where reproduction and growth would be maximized (Table 3.1). It is worth noting that, for *B. schlosseri*, CT_{min} is greater when reproduction is used as the performance trait than when growth is used (Table 3.1). This occurs because reproductive output, unlike growth, is stalled when temperatures are lower than 20°C, a trade-off commonly observed in marine invertebrates (Sebens, 2002; Sibby and Atkinson, 1994).

Because body size is a relatively straightforward parameter to measure, growth is commonly used as a performance trait. In addition, growth is regarded as a fairly accurate proxy for fitness. An organism's growth represents a net yield, which results from the difference between energy costs (metabolic cost) and benefits (ingestion/assimilation rate) (Levinton, 1983; Sanford, 2002a). Thus, aquaculture-driven research has dedicated a considerable amount of effort to understanding energetic constraints on a number of marine and freshwater species of interest (e.g. salmon and trout species). For example, Ojanguren *et al.* (2001) elegantly describe thermal performance curves for juvenile activity levels, feeding attempts and growth rates (Table 3.1).

Feeding and/or assimilation rates have also been extensively used as proxies for temperature-dependent fitness. The ecological importance of feeding rates is overarching as it not only provides an organism-specific condition index but also sheds direct light on processes that occur at higher (i.e. population and community) levels (Paine, 1966). Southward (1955a,b, 1957) studied cirral activity, a proxy for feeding rates of different intertidal barnacle species that inhabit rocky shores of the United Kingdom. He provided extensive data on the effects of temperature throughout their thermotolerance windows (Table 3.1), revealing a clear fit with a typical performance curve's shape (Fig. 3.6). Furthermore, his research draws attention to differences in performance in relation to additional sources of variability, including geographic origin of the species and populations (including the potential for thermal adaptation and acclimation) and intertidal height. For example, he showed how the species *Balanus perforatus* and *Chthamalus stellatus* exhibited higher P_{max} and T_{max} than *S. (Balanus) balanoides*, in accordance with their more southern distribution (Southward 1955a,b). Notably, as highlighted above, studies that only concentrate on feeding rates

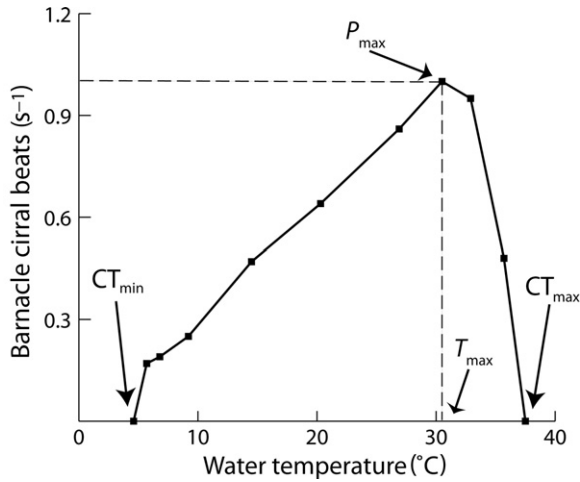


Figure 3.6 Performance curve based on barnacle cirral beat frequency, as reported by Southward (1955a,b).

may not be sufficient to explain the entire story. Although feeding rates are known to increase with temperature — presumably allowing for higher growth, reproduction and fitness — the energetic costs (metabolic rate) are known to rise exponentially with temperature as well, compromising the overall contribution of increased feeding rates shown by the organism (Levinton, 1983; Sanford, 2002a).

In sum, while many examples exist in the literature of the effects of environmental factors on traits related to fitness and performance, in relatively few cases do we have complete performance curves for marine organisms, and especially non-commercial invertebrates. Many studies have focused on extremes of temperature, or (especially in the case of aquaculture studies) on optima. This area thus represents a major gap in our knowledge, but is one that could be filled relatively easily. It is a critical area. For example, the outcome of field manipulations that compare the influence of environmental parameters on physiological or ecological performance depends entirely on where the conditions of the two sites lie on the performance curve; e.g. an increase in habitat temperature could lead to an increase in performance, a decrease in performance, or no change at all depending entirely on what combination of conditions were used (Fig. 3.5). While axiomatic in hindsight, often field manipulations fail to take such relationships into consideration or, when they do, they do not measure the conditions of the organism directly, but rather rely on proxies such as air or water temperature that may or may not accurately represent the physiological status of the organism (Figs 3.2 and 3.5).

4. INDIRECT EFFECTS OF CLIMATE CHANGE: SPECIES INTERACTIONS AND TIPPING POINTS

Key to our argument is the concept that changes in the population dynamics of one or a few species can lead to community-level phase shifts/tipping points, and that the reproductive failure or large-scale mortality of a species will not simply result in its replacement by a functional equivalent. Clearly this argument will not apply to all organisms or ecosystems. For example, there are many discussions in the literature of functional redundancy, especially in planktonic communities, and of the role of guilds. However, many examples of the important role played by a few key species have been shown, as in the case of organisms at the base of food webs such as the Antarctic pteropod (example given earlier).

The concept of keystone species, first introduced by Paine (1969), has become a cornerstone of ecological theory (Power *et al.*, 1996), and while its universality has been questioned (Strong, 1992), experimental manipulations have shown that it serves as a useful heuristic tool for examining interaction strengths within food webs (Menge *et al.*, 1994), and the disproportionately large importance that some species have in relation to their abundance, i.e. the definition of a keystone species (Power *et al.*, 1996). In some definitions, the keystone species concept is remarkably similar to that of a tipping point. For example, in his criticism of the keystone species concept, Strong (1992) defines keystone species as ‘taxa with such top-down dominance that their removal causes precipitous change in the [eco]system’.

In a now-famous experiment Paine (1974) experimentally removed the predatory seastar *Pisaster* over a period of 5 years from a rocky intertidal shore. The removal of this keystone predator resulted in the competitive dominance of the primary space occupier *M. californianus*, which then excluded over 25 other species of invertebrates and benthic algae from the shore (although, as noted by Suchanek (1992), the presence of mussel beds also increases the diversity of fauna living within the bed). Estes and Palmisano (1974) have shown that the removal of sea otters results in rapid expansion of sea urchin populations, which in turn destroy macroalgae, resulting in urchin barrens. On coral reefs, the presence of herbivores determines whether reefs undergo a phase shift from a coral-dominated reef to an algal-dominated reef (but see Dudgeon *et al.*, 2010). Hoey and Bellwood (2009) quantified rates of browsing on *Sargassum* on the Great Barrier Reef and found that despite the fact that the reef supported 50 herbivorous fish species and 6 macroalgal browsing species, a single species, *Naso unicornis* Forsskal, 1775, was responsible for ~95% of the algae removed via grazing. Thus, this single species determined to a large extent the phase transition from a coral-dominated to a macroalgal-dominated reef.

Forty years after the introduction of the term, the definition of a keystone species is still debated, and has taken on new urgency given its implications for conservation management (Clemente *et al.*, 2010; Navarrete *et al.*, 2010). The discussion over whether keystone species should receive special management status continues, and several authors have suggested that the concept be expanded to include any species that has a large impact on their assemblage, whether out of proportion to their biomass or not (Mills *et al.*, 1993; Davic, 2003). For example, in a 1993 review, Mills *et al.* described five different types of keystone species: keystone predators, keystone prey, keystone mutualists, keystone hosts and keystone modifiers. *Keystone predators* typically act by removing competitive dominants or other consumers, as described above. *Keystone prey* species affect community diversity through their impacts on the populations of their predators; via high rates of reproduction, these species are able to sustain populations of predators, thereby reducing the density of other prey species (Holt, 1977). *Keystone mutualists* are species that are critical to mutualistic relationships, e.g. pollinators and seed dispersers. *Keystone hosts* are the organisms that in turn support those pollinators and dispersers, e.g. plants. While Mills *et al.* (1993) did not list any marine examples of these latter two categories, zooxanthellae and their coral hosts may potentially lend themselves to these definitions.

Finally, Mills *et al.* (1993) defined *keystone modifiers* as species that significantly altered the physical habitat without necessarily having any trophic relationship with other species. The archetypical example given was that of North American beavers, which through the creation of dams flood the landscape, thereby impacting all other members of the assemblage. Jones *et al.* (1994) expanded on this latter definition to explore the concept of organisms as physical ecosystem engineers. Ecosystem engineers physically modify, maintain or create habitat, and in doing so directly or indirectly control the availability of resources to other species. Thus, for example, ecosystem engineers such as trees, corals, tube worms and bed-forming animals such as mussels and oysters all create living space for other organisms. Suchanek (1992) noted 135 species living in beds of the mussel *M. californianus*. Some polychaete species (e.g. *Diopatra neapolitana* Delle Chiaje, 1841) can build large emergent tubes that can alter flow regimes, stabilize sediment, and drive patterns of biodiversity by providing refugia for other species from predation (Woodin, 1981). Other species (e.g. *Arenicola marina* Linnaeus, 1758) are bioturbators that create disturbance that leads to decreases in biodiversity (Berke *et al.*, 2010).

In all of these examples, one species has a large effect on the ecological community, and thus any sudden change in levels of physiological stress, reproduction or mortality that affect the behaviour and/or population dynamics of those species is likely to have a cascading ecological impact (Connell *et al.*, 2011; Kordas *et al.*, 2011). In many cases these impacts are likely to exhibit a threshold effect as well. Below critical population

densities, populations can exhibit an Allee effect (depensation) in which negative rates of *per capita* growth begin to occur (Stoner and Ray-Culp, 2000). For example, below critical densities spawning success of urchins has been shown to decline (Levitan *et al.*, 1992). Recent work has shown that small populations, especially those at the edge of species ranges ('frayed edges'), are more highly susceptible to environmental change (i.e. less physiologically resilient) due to lower genetic variance (Pearson *et al.*, 2009). Results such as these are worrisome, because they suggest that thresholds may occur even more rapidly once some minimum threshold in genetic variance, and hence a lower adaptive capacity, is surpassed. Cumulatively these studies point to the need for a better understanding not only of the direct physiological effects of climate change but also the indirect effects on species interactions.

Multiple studies have examined the indirect effects of changes in climate-related factors on species interactions (Poloczanska *et al.*, 2008; Yamane and Gilman, 2009; Connell *et al.*, 2011; Kordas *et al.*, 2011). Wethey (1984) experimentally altered the outcome of competitive interactions by shading two species of intertidal barnacles, demonstrating that the dominant competitor was restricted from the more physiologically challenging high intertidal zone by thermal and/or desiccation stress. Schneider *et al.* (2010) compared the survival of two species of mussels (*Mytilus trossulus* Gould, 1850, and *Mytilus galloprovincialis* Lamarck, 1819) under varying conditions of aerial exposure and food availability and reported differential survival under stressful aerial conditions, suggesting a role of environmental stress in driving the distribution of these two species, one of which (*M. galloprovincialis*) is an invasive. Sanford (1999, 2002a) showed that rates of predation by the seastar *P. ochraceus* on the intertidal mussel *M. californianus* were positively correlated with water temperature. Pincebourde *et al.* (2008) expanded upon this work and showed that the aerial body temperature of *Pisaster* also affected feeding rates. Following short (1–2 day) exposures to elevated temperatures, increasing aerial body temperatures led to higher feeding rates. However, following longer (8 day) exposures to temperatures that were high yet still realistic when compared to what was observed in the field, feeding rates decreased by up to 40%, and led to decreases in seastar growth (Pincebourde *et al.*, 2008).



5. PUTTING THE PIECES TOGETHER: WHERE DO WE GO FROM HERE?

Given the complicated interactions between organisms and their physical environments, the physiological mechanisms by which environmental factors drive organism behaviour, fitness and survival, and the indirect effects of

these impacts on species interactions, is there any hope for a mechanistic framework? Multiple models have been proposed to explore the relationship between abiotic stressors and species interactions. For example, consumer stress models posit that top predators are more affected by physiological stress than are their prey (Menge and Sutherland, 1987). In contrast, prey stress models suggest that prey experience higher levels of physiological stress than do their predators (Menge and Olson, 1990). A quantitative understanding of relative physiological stress levels of predator and prey under both normal and extreme field conditions is thus vital to the application of these theories (Petes *et al.*, 2008). Critically, the concepts presented in this chapter suggest not only that patterns in the field may be more complex than anticipated, but they also may be highly context dependent. That is, a site that is physiologically stressful for one species may not necessarily be so for another, for several reasons (Fig. 3.7). For example, a prey species may be physiologically more stressed than its predator because the predator has a thermal performance curve with higher optimum and lethal temperature limits. However, a prey species may also be more stressed than its predator simply because, under the same environmental conditions, the predator maintains a lower body temperature (Fig. 3.7). This appears to be the case with the intertidal seastar *P. ochraceus* and its mussel prey *M. californianus* (Petes *et al.*, 2008). Largely because of its wet surface and large thermal inertia (Pincebourde *et al.*, 2009), *Pisaster* appears to maintain a temperature that is either the same or lower than that of its prey (Broitman *et al.*, 2009). Even if the two species have similar performance curves (e.g., *Pisaster* and *Mytilus* appear to have similar lethal limits: Pincebourde *et al.*, 2008; Denny *et al.*, 2011), they will experience very different levels of stress under identical field conditions simply because the predator is able to maintain a lower body temperature. Thus, neither measurements of performance curves nor measurements of body temperature alone are enough to determine relative (or even absolute) stress levels in the field; and certainly measurements of habitat alone are insufficient.

Predicting physiological stress levels under field conditions is of course no simple matter due to potential interactions between multiple stressors. Crain *et al.* (2008) reviewed 171 studies that manipulated two or more stressors in marine ecosystems. Their meta-analysis showed an overall synergistic interaction effect, suggesting that the cumulative effects of multiple stressors are likely to be worse than expected based on the independent impacts of each stressor. Moreover, they found that cumulative effects could be additive, synergistic or antagonistic. Thus, in some cases, stressors either ameliorated one another, or one stressor had such a large effect the addition of a second stressor had no additional impact.

Still, some studies have shown that when there is an overwhelming effect of one stressor, explicit predictions can be made that can then be tested under field conditions. Hofmann *et al.* (2010) outlined an approach that links differential susceptibility to OA, including physiological plasticity,

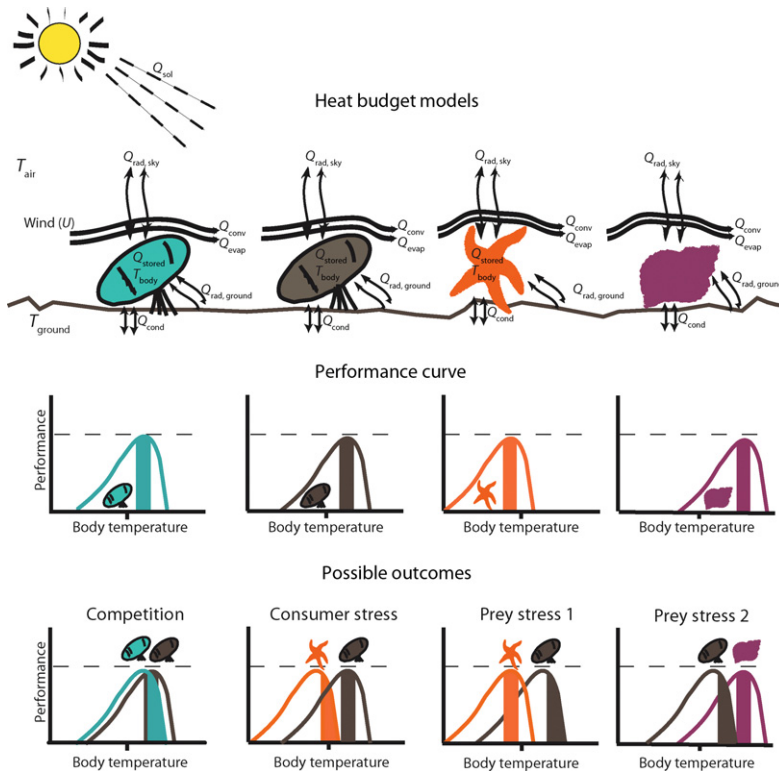


Figure 3.7 Translation of changes in the physical environment into ecological responses for two prey species (*Mytilus* spp.) and two consumers (*Pisaster* and *Nucella*). Because characteristics such as the size, morphology, colour and mass of an organism drives rates of heat flux, two organisms exposed to identical microclimates can have very different body temperatures. Each organism also has a unique physiological response to body temperature (thermal performance curve). The relative performance of interacting organisms can then influence the outcome of competitive interactions, or of rates of predation. Thus, for example, under elevated temperatures ‘prey stress’ can occur either because in any particular environment, the predator is able to maintain a lower body temperature, as appears to be the case between *Pisaster* and *Mytilus* (‘Prey Stress 1’), and/or because of a higher physiological tolerance of the consumer to temperature (‘Prey Stress 2’).

to spatially explicit measurements of ocean pH in order to assess global patterns of calcification. Terrestrial ecologists have long used mechanistic heat budget models to generate temporally and spatially explicit maps of body temperature that can then be compared against known tolerance limits derived from controlled laboratory and field studies (Kearney and Porter, 2009). More recently these approaches have been applied to

intertidal ecosystems (Gilman *et al.*, 2006; Kearney *et al.*, 2010; Denny *et al.*, 2011; Helmuth *et al.*, 2011). Such studies hold a distinct advantage over correlative ('climate envelope') models in that they can potentially incorporate local adaptation (Kuo and Sanford, 2009) and acclimation (Somero, 2010); however, they also are much more time- and data intensive.

Most recently, biophysical models have been connected to Dynamic Energy Budget (DEB) models (Kooijman, 2009) as a means of accurately estimating the effects of changing levels of food and temperature on growth, reproductive output and survival (Kearney *et al.*, 2010; Sará *et al.*, 2011). Unlike other conceptual models, DEB models recognize that organisms live, as the name implies, in a dynamic environment. Thus, for example, Fig. 3.4 implies that it is a simple matter to define performance level at any given temperature (or combination of food and temperature, etc.). In reality, however, except for organisms in environments such as the deep sea or polar waters, organisms seldom live at a fixed temperature. Even in relatively thermally stable environments such as coral reefs, water temperatures fluctuate by up to several degrees due to solar heating of surface waters (Leichter *et al.*, 2006; Castillo and Lima, 2010). Fluctuations in body temperature of 25°C or more are not uncommon in intertidal environments (Finke *et al.*, 2009; Marshall *et al.*, 2010). In reality, therefore, organismal performance changes throughout the course of the day as body temperature changes, although mobile organisms can potentially ameliorate much of that variability through microhabitat selection. As a corollary, the cumulative effects of thermal variability have been shown to significantly affect an organism's long-term performance. For example, Sanford (2002b) compared feeding and growth rates of the intertidal predators *P. ochraceus* and *Nucella canaliculata* Duclos, 1832, held in tanks at three temperature treatments: constant 'cold' (9°C), constant 'warm' (12°C), and a 14-day fluctuating regime (9–12°C) simulating recurrent upwelling conditions. His experiments revealed greater growth under a fluctuating thermal environment, which led him to speculate that a continuous displacement in body temperature around the $T_{\max}$ of a thermal performance curve would explain such results. Importantly, as theoretical models (Katz *et al.*, 2005) and empirical studies (Easterling *et al.*, 2000) have demonstrated that climate change involves increases in temperature variability, models designed to forecast organisms' response to climate change need to explicitly consider scenarios with different thermal amplitudes (Folguera *et al.*, 2009). The use of a single temperature to represent the physiological state of an animal over the course of a day (or even longer time period) is therefore generally untenable, especially when average values are used. DEB bypasses such limitations through a dynamic approach.

Kearney *et al.* (2010) successfully combined a biophysical heat budget model with a DEB model to study the effects of aerial and aquatic temperature on the intertidal mussel *M. californianus*. While they did not use the model to explore potential future effects of climate change, they did demonstrate the efficacy of the approach in predicting patterns of reproductive output and growth. Thus, the model could easily be combined with projections of future climate scenarios to predict the conditions under which this major space occupier would be most likely to experience mortality and/or reproductive failure, thus leading to a major phase shift in the intertidal ecosystem. Alternatively, it could be used as part of a sensitivity analysis to determine which environmental parameters are most likely to impact this key species (Helmuth *et al.*, 2011).

Mechanistic forecasting approaches such as these thus hold considerable promise, especially if and when they can begin to incorporate indirect effects such as predation (Sanford, 2002a; Petes *et al.*, 2008; Pincebourde *et al.*, 2008; Yamane and Gilman, 2009). However, their ability to predict such complex interactions can only be assessed empirically; and because it is imperative that we test such models now rather than wait to see what will happen under future climate scenarios, our best option is thus to use nowcasting and hindcasting as hypothesis-testing frameworks (Helmuth *et al.*, 2006; Wethey and Woodin, 2008). Specifically, using our understanding of the sensitivity of organisms to changes in environmental parameters, we can develop much more sophisticated predictions of the likely effects of changes in the physical environment that can then be tested using experimental manipulations (Firth and Williams, 2009). The combination of both organismal- and sub-organismal scale measurements and models with studies at population and community scales will provide a much more comprehensive view of the drivers of ecological thresholds than will simple correlations between environmental change and community responses. More to the point, an understanding of the world as viewed by the organisms we study will place us in a much better stead if we are to have any hope of predicting, and hopefully averting, some of the most severe impacts of anthropogenic change in coming decades.

ACKNOWLEDGEMENTS

The authors were supported by grants from NSF (OCE-0926581) and NASA (NNX07AF20G). We would particularly like to thank Michael Lesser for his help and patience throughout the process of writing this manuscript, and Andrés Monaco for assistance in the creation of figures. We are grateful to Carol Blanchette, Bernardo Broitman, Nicholas Burnett, Nicholas Colvard, Jerry Hilbish, Josie Iacarella, Michael Kearney, Allison Matzelle, Laura Petes, Gianluca Sará, Allison Smith, David Wethey and Sally Woodin for the many discussions that contributed significantly to the ideas presented in this manuscript.

REFERENCES

- Agrawala, S., Broad, K. and Guston, D. H. (2001). Integrating climate forecasts and societal decision making: Challenges to an emergent boundary organization. *Science Technology and Human Values* **26**, 454–477.
- Angilletta, M. J. (2009). *Thermal Adaptation: A Theoretical and Empirical Synthesis*. Oxford University Press, New York.
- Angilletta, M. J., Niewiarowski, P. H. and Navas, C. A. (2002). The evolution of thermal physiology in ectotherms. *Journal of Thermal Biology* **27**, 249–268.
- Angilletta, M. J., Wilson, R. S., Navas, C. A. and James, R. S. (2003). Tradeoffs and the evolution of thermal reaction norms. *Trends in Ecology & Evolution* **18**, 234–240.
- Baumont, L. J. and Hughes, L. (2002). Potential changes in the distributions of latitudinally restricted Australian butterfly species in response to climate change. *Global Change Biology* **8**, 954–971.
- Bell, E. C. (1995). Environmental and morphological influences on thallus temperature and desiccation of the intertidal alga *Mastocarpus papillatus* Kützting. *Journal of Experimental Marine Biology and Ecology* **191**, 29–55.
- Bertness, M. D. and Schneider, D. E. (1976). Temperature relations of Puget Sound thalids in reference to their intertidal distribution. *Veliger* **19**, 47–58.
- Berke, S. K. (2010). Functional groups of ecosystem engineers: A proposed classification with comments on current issues. *Integrative and Comparative Biology* **50**, 147–157.
- Beukema, J. J., Dekker, R. and Jansen, J. M. (2009). Some like it cold: Populations of the tellinid bivalve *Macoma balthica* (L.) suffer in various ways from a warming climate. *Marine Ecology Progress Series* **384**, 135–145.
- Blanchette, C. A., Miner, M. M., Raimondi, P. T., Lohse, D., Heady, K. E. K. and Broitman, B. R. (2008). Biogeographical patterns of rocky intertidal communities along the Pacific coast of North America. *Journal of Biogeography* **35**, 1593–1607.
- Boller, M. L. and Carrington, E. (2007). Interspecific comparison of hydrodynamic performance and structural properties among intertidal macroalgae. *Journal of Experimental Biology* **210**, 1874–1884.
- Brett, J. R. (1971). Energetic responses of salmon to temperature – study of some thermal relations in physiology and freshwater ecology of sockeye salmon (*Oncorhynchus nerka*). *American Zoologist* **11**, 99–113.
- Briske, D. D., Washington-Allen, R. A., Johnson, C. R., Lockwood, J. A., Lockwood, D. R., Stringham, T. K. and Shugart, H. H. (2010). Catastrophic thresholds: A synthesis of concepts, perspectives, and applications. *Ecology and Society* **15**, 37.
- Broitman, B. R., Szathmary, P. L., Mislán, K. A. S., Blanchette, C. A. and Helmuth, B. (2009). Predator–prey interactions under climate change: The importance of habitat vs body temperature. *Oikos* **118**, 219–224.
- Brunetti, R., Beghi, L., Bressan, M. and Marin, M. G. (1980). Combined effects of temperature and salinity on colonies of *Botryllus schlosseri* and *Botryllodes leachi* (Ascidacea) from the Venetian Lagoon. *Marine Ecology Progress Series* **2**, 303–314.
- Carrington, E. (2002). Seasonal variation in the attachment strength of blue mussels: Causes and consequences. *Limnology and Oceanography* **47**, 1723–1733.
- Carrington, E., Moeser, G. M., Dimond, J., Mello, J. J. and Boller, M. L. (2009). Seasonal disturbance to mussel beds: Field test of a mechanistic model predicting wave dislodgment. *Limnology and Oceanography* **54**, 978–986.
- Casini, M., Hjelm, J., Molinero, J.-C., Lövgren, J., Cardinale, M., Bartolino, V., Belgrano, A. and Kornilovs, G. (2009). Trophic cascades promote threshold-like shifts in pelagic marine ecosystems. *Proceedings of the National Academy of Sciences of the United States of America* **106**, 197–202.
- Castillo, K. D. and Lima, F. P. (2010). Comparison of in situ and satellite-derived (MODIS-Aqua/Terra) methods for assessing temperatures on coral reefs. *Limnology and Oceanography: Methods* **8**, 107–117.

- CCSP (2008). The effects of climate change on agriculture, land resources, water resources, and biodiversity. In *A Report by the U.S. Climate Change Science Program and the Subcommittee on Global Change Research* (A.J.P. Backlund, D. Schimel, J. Hatfield, K. Boote, P. Fay, L. Hahn, C. Izaurralde, B.A. Kimball, T. Mader, J. Morgan, D. Ort, W. Polley, A. Thomson, D. Wolfe, M. Ryan, S. Archer, R. Birdsey, C. Dahm, L. Heath, J. Hicke, D. Hollinger, T. Huxman, G. Okin, R. Oren, J. Randerson, W. Schlesinger, D. Lettenmaier, D. Major, L. Poff, S. Running, L. Hansen, D. Inouye, B.P. Kelly, L. Meyerson, B. Peterson and R. Shaw, eds), pp. 332. US Environmental Protection Agency, Washington, DC.
- CCSP (2009). Thresholds of climate change in ecosystems. In *A Report by the U.S. Climate Change Science Program and the Subcommittee on Global Change Research* (D.B. Fagre, C.W. Charles, C.D. Allen, C. Birkeland, F.S. Chapin III, P.M. Groffman, G.R. Guntenspergen, A.K. Knapp, A.D. McGuire, P.J. Mulholland, D.P.C. Peters, D.D. Roby and G. Sugihara, eds), pp. 156. USG Survey, Reston, VA.
- Chown, S. L. and Gaston, K. J. (2008). Macrophysiology for a changing world. *Proceedings of the Royal Society of London Series B* **275**, 1469–1478.
- Chown, S. L., Gaston, K. J. and Robinson, D. (2004). Macrophysiology: Large-scale patterns in physiological traits and their ecological implications. *Functional Ecology* **18**, 159–167.
- Clemente, S., Hernández, J. C., Rodríguez, A. and Brito, A. (2010). Identifying keystone predators and the importance of preserving functional diversity in sublittoral rocky-bottom areas. *Marine Ecology Progress Series* **413**, 55–67.
- Connell, S. D., Russell, B. D. and Irving, A. D. (2011). Can strong consumer and producer effects be reconciled to better forecast ‘catastrophic’ phase-shifts in marine ecosystems? *Journal of Experimental Marine Biology and Ecology* **400**, 296–301.
- Crain, C. M., Kroeker, K. and Halpern, B. S. (2008). Interactive and cumulative effects of multiple human stressors in marine systems. *Ecology Letters* **11**, 1304–1315.
- Crimmins, S. M., Dobrowski, S. Z., Greenberg, J. A., Abatzoglou, J. T. and Mynsberge, A. R. (2011). Changes in climatic water balance drive downhill shifts in plant species’ optimum elevations. *Science* **331**, 324–327.
- Davic, R. D. (2003). Linking keystone species and functional groups: A new operational definition of the keystone species concept. *Conservation Ecology* **7**, r11.
- Davis, A. J., Jenkinson, L. S., Lawton, J. H., Shorrocks, B. and Wood, S. (1998a). Making mistakes when predicting shifts in species range in response to global warming. *Nature* **391**, 783–786.
- Davis, A. J., Lawton, J. H., Shorrocks, B. and Jenkinson, L. S. (1998b). Individualistic species responses invalidate simple physiological models of community dynamics under global environmental change. *Journal of Animal Ecology* **67**, 600–612.
- Denny, M. and Gaylord, B. (2010). Marine ecomechanics. *Annual Review of Marine Science* **2**, 89–114.
- Denny, M. W. and Helmuth, B. (2009). Confronting the physiological bottleneck: A challenge from ecomechanics. *Integrative and Comparative Biology* **49**, 197–201.
- Denny, M. W., Helmuth, B., Leonard, G. H., Harley, C. D. G., Hunt, L. J. H. and Nelson, E. K. (2004). Quantifying scale in ecology: Lessons from a wave-swept shore. *Ecological Monographs* **74**, 513–532.
- Denny, M. W., Dowd, W. W., Bilir, L. and Mach, K. J. (2011). Spreading the risk: Small-scale body temperature variation among intertidal organisms and its implications for species persistence. *Journal of Experimental Marine Biology and Ecology* **400**, 175–190.
- Dewitt, C. B. and Friedman, R. M. (1979). Significance of skewness in ectotherm thermoregulation. *American Zoologist* **19**, 195–209.
- Dudgeon, S. R., Aronson, R. B., Bruno, J. F. and Pecht, W. F. (2010). Phase shifts and stable states on coral reefs. *Marine Ecology Progress Series* **413**, 201–216.

- Easterling, D. R., Meehl, G. A., Parmesan, C., Changnon, S. A., Karl, T. R. and Mearns, L. O. (2000). Climate extremes: Observations, modeling, and impacts. *Science* **289**, 2068–2074.
- Epelbaum, A., Herborg, L. M., Therriault, T. W. and Pearce, C. M. (2009). Temperature and salinity effects on growth, survival, reproduction, and potential distribution of two non-indigenous botryllid ascidians in British Columbia. *Journal of Experimental Marine Biology and Ecology* **369**, 43–52.
- Estes, J. A. and Palmisano, J. F. (1974). Sea otters - Their role in structuring seashore communities. *Science* **185**, 1058–1060.
- Fabrizius, K. E. (2006). Effects of irradiance, flow, and colony pigmentation on the temperature microenvironment around corals: Implications for coral bleaching? *Limnology and Oceanography* **51**, 30–37.
- Fabry, V. J. (2008). Marine calcifiers in a high-CO₂ ocean. *Science* **320**, 1020–1022.
- Finelli, C. M., Helmuth, B. S. T., Pentcheff, N. D. and Wethey, D. S. (2006). Water flow controls oxygen transport and photosynthesis in corals: Potential links to coral bleaching. *Coral Reefs* **25**, 47–57.
- Finelli, C. M., Helmuth, B. S., Pentcheff, N. D. and Wethey, D. S. (2007). Intracolony variability in photosynthesis by corals is affected by water flow: Role of oxygen flux. *Marine Ecology Progress Series* **349**, 103–110.
- Finke, G. R., Navarrete, S. A. and Bozinovic, F. (2007). Tidal regimes of temperate coasts and their influences on aerial exposure for intertidal organisms. *Marine Ecology Progress Series* **343**, 57–62.
- Finke, G. R., Bozinovic, F. and Navarrete, S. A. (2009). A mechanistic model to study the thermal ecology of a southeastern Pacific dominant intertidal mussel and implications for climate change. *Physiological and Biochemical Zoology* **82**, 303–313.
- Firth, L. B. and Williams, G. A. (2009). The influence of multiple environmental stressors on the limpet *Cellana toreuma* during the summer monsoon season in Hong Kong. *Journal of Experimental Marine Biology and Ecology* **375**, 70–75.
- Folguera, G., Bastías, D. A. and Bozinovic, F. (2009). Impact of experimental thermal amplitude on ectotherm performance: Adaptation to climate change variability? *Comparative Biochemistry and Physiology - Part A, Molecular & Integrative Physiology* **154**, 389–393.
- Forchhammer, M. C. and Post, E. (2004). Using large-scale climate indices in climate change ecology studies. *Population Ecology* **46**, 1–12.
- Frederich, M. and Pörtner, H. O. (2000). Oxygen limitation of thermal tolerance defined by cardiac and ventilatory performance in spider crab, *Maja squinado*. *American Journal of Physiology - Regulatory Integrative and Comparative Physiology* **279**, R1531–R1538.
- Gedan, K. B. and Bertness, M. D. (2009). Experimental warming causes rapid loss of plant diversity in New England salt marshes. *Ecology Letters* **12**, 842–848.
- Gilman, S. E., Wethey, D. S. and Helmuth, B. (2006). Variation in the sensitivity of organismal body temperature to climate change over local and geographic scales. *Proceedings of the National Academy of Sciences of the United States of America* **103**, 9560–9565.
- Goodwin, J. and Dahlstrom, M. F. (2011). Good reasons for trusting climate science communication. American Meteorological Society (abstract), Washington, DC.
- Gouhier, T. C., Guichard, F. and Menge, B. A. (2010). Ecological processes can synchronize marine population dynamics over continental scales. *Proceedings of the National Academy of Sciences of the United States of America* **107**, 8281–8286.
- Gray, J. (1923). The mechanism of ciliary movement. III. The effect of temperature. *Proceedings of the Royal Society of London Series B - Containing Papers of a Biological Character* **95**, 6–15.

- Hallett, T. B., Coulson, T., Pilkington, J. G., Clutton-Brock, T. H., Pemberton, J. M. and Grenfell, B. T. (2004). Why large-scale climate indices seem to predict ecological processes better than local weather. *Nature* **430**, 71–75.
- Hare, S. R. and Mantua, N. J. (2000). Empirical evidence for North Pacific regime shifts in 1977 and 1989. *Progress in Oceanography* **47**, 103–145.
- Harley, C. D. G. (2008). Tidal dynamics, topographic orientation, and temperature-mediated mass mortalities on rocky shores. *Marine Ecology Progress Series* **371**, 37–46.
- Harley, C. D. G. and Paine, R. T. (2009). Contingencies and compounded rare perturbations dictate sudden distributional shifts during periods of gradual climate change. *Proceedings of the National Academy of Sciences of the United States of America* **106**, 11172–11176.
- Helmuth, B. (2002). How do we measure the environment? Linking intertidal thermal physiology and ecology through biophysics. *Integrative and Comparative Biology* **42**, 837–845.
- Helmuth, B. (2009). From cells to coastlines: How can we use physiology to forecast the impacts of climate change? *Journal of Experimental Biology* **212**, 753–760.
- Helmuth, B. and Denny, M. W. (2003). Predicting wave exposure in the rocky intertidal zone: Do bigger waves always lead to larger forces? *Limnology and Oceanography* **48**, 1338–1345.
- Helmuth, B. S., Harley, C. D. G., Halpin, P., O'Donnell, M., Hofmann, G. E. and Blanchette, C. (2002). Climate change and latitudinal patterns of intertidal thermal stress. *Science* **298**, 1015–1017.
- Helmuth, B., Carrington, E. and Kingsolver, J. G. (2005). Biophysics, physiological ecology, and climate change: Does mechanism matter? *Annual Review of Physiology* **67**, 177–201.
- Helmuth, B., Broitman, B. R., Blanchette, C. A., Gilman, S., Halpin, P., Harley, C. D. G., O'Donnell, M. J., Hofmann, G. E., Menge, B. and Strickland, D. (2006). Mosaic patterns of thermal stress in the rocky intertidal zone: Implications for climate change. *Ecological Monographs* **76**, 461–479.
- Helmuth, B., Broitman, B. R., Yamane, L., Gilman, S. E., Mach, K., Mislan, K. A. S. and Denny, M. W. (2010). Organismal climatology: Analyzing environmental variability at scales relevant to physiological stress. *Journal of Experimental Biology* **213**, 995–1003.
- Helmuth, B., Yamane, L., Lalwani, S., Matzelle, A., Tockstein, A. and Gao, N. (2011). Hidden signals of climate change in intertidal ecosystems: What (not) to expect when you are expecting. *Journal of Experimental Marine Biology and Ecology* **400**, 191–199.
- Hewitt, J. E. and Thrush, S. F. (2010). Empirical evidence of an approaching alternate state produced by intrinsic community dynamics, climatic variability and management actions. *Marine Ecology Progress Series* **413**, 267–276.
- Highsmith, R. C., Riggs, A. C. and D'Antonio, C. M. (1980). Survival of hurricane-generated coral fragments and a disturbance model of reef calcification/growth rates. *Oecologia* **46**, 322–329.
- Hilbert, D. W. (2002). Non-linear systems. In “Encyclopedia of Global Environmental Change, The Earth System: Biological and Ecological Dimensions of Global Environmental Change” (T. Munn, H. A. Mooney and J. G. Canadell, eds), vol. II, pp. 450–455. John Wiley & Sons, Chichester.
- Hoegh-Guldberg, O. and Bruno, J. F. (2010). The impact of climate change on the world's marine ecosystems. *Science* **328**, 1523–1528.
- Hoegh-Guldberg, O., Mumby, P. J., Hooten, A. J., Steneck, R. S., Greenfield, P., Gomez, E., Harvell, C. D., Sale, P. F., Edwards, A. J. Caldeira, K., et al. (2007). Coral reefs under rapid climate change and ocean acidification. *Science* **318**, 1737–1742.
- Hoey, A. S. and Bellwood, D. R. (2009). Limited functional redundancy in a high diversity system: Single species dominates key ecological process on coral reefs. *Ecosystems* **12**, 1316–1328.

- Hoffman, J. (2003). Temperate marine. In "Buying Time: A User's Manual for Building Resistance and Resilience to Climate Change in Natural Systems" (L. J. Hansen, J. L. Biringer and J. R. Hoffman, eds), pp. 123–155.
- Hofmann, G. E., Barry, J. P., Edmunds, P. J., Gates, R. D., Hutchins, D. A., Klinger, T. and Sewell, M. A. (2010). The effect of ocean acidification on calcifying organisms in marine ecosystems: An organism-to-ecosystem perspective. *Annual Review of Ecology Evolution and Systematics* **41**, 127–147.
- Holt, R. D. (1977). Predation, apparent competition, and the structure of prey communities. *Theoretical Population Biology* **12**, 197–229.
- Holtmeier, F.-K. and Broll, G. (2005). Sensitivity and response of northern hemisphere altitudinal and polar treelines to environmental change at landscape and local scales. *Global Ecology and Biogeography* **14**, 395–410.
- Huey, R. B. and Stevenson, R. D. (1979). Integrating thermal physiology and ecology of ectotherms: A discussion of approaches. *American Zoologist* **19**, 357–366.
- Hunt, B. P. V., Pakhomov, E. A., Hosie, G. W., Siegel, V., Ward, P. and Bernard, K. (2008). Pteropods in Southern Ocean ecosystems. *Progress in Oceanography* **78**, 193–221.
- Hutchins, D. A., Fu, F.-X., Zhang, Y., Warner, M. E., Feng, Y., Portune, K., Bernhardt, P. W. and Mulholland, M. R. (2007). CO₂ control of *Trichodesmium* N₂ fixation, photosynthesis, growth rates, and elemental ratios: Implications for past, present, and future ocean biogeochemistry. *Limnology and Oceanography* **52**, 1293–1304.
- Idjadi, J., Lee, S., Bruno, J., Precht, W., Allen-Requa, L. and Edmunds, P. (2006). Rapid phase-shift reversal on a Jamaican coral reef. *Coral Reefs* **25**, 209–211.
- Jimenez, I. M., Kuhl, M., Larkum, A. W. D. and Ralph, P. J. (2008). Heat budget and thermal microenvironment of shallow-water corals: Do massive corals get warmer than branching corals? *Limnology and Oceanography* **53**, 1548–1561.
- Jones, C. G., Lawton, J. H. and Shachak, M. (1994). Organisms as ecosystem engineers. *Oikos* **69**, 373–386.
- Kassahn, K. S., Crozier, R. H., Pörtner, H. O. and Caley, M. J. (2009). Animal performance and stress: Responses and tolerance limits at different levels of biological organisation. *Biological Reviews* **84**, 277–292.
- Katz, R. W., Brush, G. S. and Parlange, M. P. (2005). Statistics of extremes: Modeling ecological disturbances. *Ecology* **36**, 1124–1134.
- Kearney, M. (2006). Habitat, environment and niche: What are we modelling? *Oikos* **115**, 186–191.
- Kearney, M. and Porter, W. (2009). Mechanistic niche modelling: Combining physiological and spatial data to predict species ranges. *Ecology Letters* **12**, 334–350.
- Kearney, M., Phillips, B. L., Tracy, C. R., Christian, K. A., Betts, G. and Porter, W. P. (2008). Modelling species distributions without using species distributions: The cane toad in Australia under current and future climates. *Ecography* **31**, 423–434.
- Kearney, M., Simpson, S. J., Raubenheimer, D. and Helmuth, B. (2010). Modelling the ecological niche from functional traits. *Philosophical Transactions of the Royal Society of London Series B* **365**, 3469–3483.
- Kingsolver, J. G., Izem, R. and Ragland, G. J. (2004). Plasticity of size and growth in fluctuating thermal environments: Comparing reaction norms and performance curves. *Integrative and Comparative Biology* **44**, 450–460.
- Kooijman, S. A. L. M. (2009). *Dynamic Energy Budget Theory for Metabolic Organisation*. Cambridge University Press, Cambridge.
- Kordas, R. L., Harley, C. D. G. and O'Connor, M. I. (2011). Community ecology in a warming world: The influence of temperature on interspecific interactions in marine systems. *Journal of Experimental Marine Biology and Ecology* **400**, 218–226.

- Kuo, E. S. L. and Sanford, E. (2009). Geographic variation in the upper thermal limits of an intertidal snail: Implications for climate envelope models. *Marine Ecology Progress Series* **388**, 137–146.
- Lambert, G. (2005). Ecology and natural history of the protochordates. *Canadian Journal of Zoology* **83**, 34–50.
- Leichter, J. J., Helmuth, B. and Fischer, A. M. (2006). Variation beneath the surface: Quantifying complex thermal environments on coral reefs in the Caribbean, Bahamas and Florida. *Journal of Marine Research* **64**, 563–588.
- Lenihan, H. S., Adjerdoud, M., Kotchen, M. J., Hensch, J. L. and Nakamura, T. (2008). Reef structure regulates small-scale spatial variation in coral bleaching. *Marine Ecology Progress Series* **370**, 127–141.
- Lesser, M. P. (1996). Elevated temperatures and ultraviolet radiation cause oxidative stress and inhibit photosynthesis in symbiotic dinoflagellates. *Limnology and Oceanography* **41**, 271–283.
- Lesser, M. P. (1997). Oxidative stress causes coral bleaching during exposure to elevated temperatures. *Coral Reefs* **16**, 187–192.
- Lesser, M. P. (2004). Experimental biology of coral reef ecosystems. *Journal of Experimental Marine Biology and Ecology* **300**, 217–252.
- Lesser, M. P., Weis, V. M., Patterson, M. R. and Jokiel, P. L. (1994). Effects of morphology and water motion on carbon delivery and productivity in the reef coral, *Pocillopora damicornis* (Linnaeus): Diffusion barriers, inorganic carbon limitation, and biochemical plasticity. *Journal of Experimental Marine Biology and Ecology* **178**, 153–179.
- Levinton, J. S. (1983). The latitudinal compensation hypothesis: Growth data and a model of latitudinal growth differentiation based upon energy budgets I. Interspecific comparison of *Ophryotrocha* (Polychaeta: Dorvilleidae). *Biological Bulletin* **165**, 686–698.
- Levitan, D. R., Sewell, M. A. and Chia, F.-S. (1992). How distribution and abundance influence fertilization success in the sea urchin *Strongylocentrotus franciscanus*. *Ecology* **73**, 248–254.
- Littler, M. M. and Littler, D. S. (2007). Assessment of coral reefs using herbivory/nutrient assays and indicator groups of benthic primary producers: A critical synthesis, proposed protocols, and critique of management strategies. *Aquatic Conservation: Marine and Freshwater Ecosystems* **17**, 195–215.
- Ludwig, D., Mangel, M. and Haddad, B. (2001). Ecology, conservation, and public policy. *Annual Review of Ecology and Systematics* **32**, 481–517.
- Marshall, D. J., McQuaid, C. D. and Williams, G. A. (2010). Non-climatic thermal adaptation: Implications for species' responses to climate warming. *Biology Letters* **6**, 669–673.
- McNeil, B. I. and Matear, R. J. (2008). Southern Ocean acidification: A tipping point at 450-ppm atmospheric CO₂. *Proceedings of the National Academy of Sciences of the United States of America* **105**, 18860–18864.
- Menge, B. A. and Olson, A. M. (1990). Role of scale and environmental factors in regulation of community structure. *Trends in Ecology and Evolution* **5**, 52–57.
- Menge, B. A. and Sutherland, J. P. (1987). Community regulation: Variation in disturbance, competition, and predation in relation to environmental stress and recruitment. *The American Naturalist* **130**, 730–757.
- Menge, B. A., Berlow, E. L., Blanchette, C. A., Navarrete, S. A. and Yamada, S. (1994). The keystone species concept: Variation in interaction strength in a rocky intertidal habitat. *Ecological Monographs* **64**, 249–286.
- Millennium Ecosystem Assessment, (2005). *Ecosystems and Well-Being: Opportunities and Challenges for Business and Industry*. World Resources Institute, Washington, DC, pp. 29.
- Mills, L. S., Soulé, M. E. and Doak, D. F. (1993). The keystone-species concept in ecology and conservation. *Bioscience* **43**, 219–224.

- Mislan, K. A. S., Wetthey, D. S. and Helmuth, B. (2009). When to worry about the weather: Role of tidal cycle in determining patterns of risk in intertidal ecosystems. *Global Change Biology* **15**, 3056–3065.
- Mitchell, N. J., Kearney, M. R., Nelson, N. J. and Porter, W. P. (2008). Predicting the fate of a living fossil: How will global warming affect sex determination and hatching phenology in tuatara? *Proceedings of the Royal Society of London Series B* **275**, 2185–2193.
- Mumby, P. J., Iglesias-Prieto, R., Hooten, A. J., Sale, P. F., Hoegh-Guldberg, O., Edwards, A. J., Harvell, C. D., Gomez, E. D., Knowlton, N., Hatzitolos, M. E., Kyewalyanga, M. S. and Muthiga, N. (2011). Revisiting climate thresholds and ecosystem collapse. *Frontiers in Ecology and the Environment* **9**, 94–96.
- Nakamura, T. and van Woesik, R. (2001). Water-flow rates and passive diffusion partially explain differential survival of corals during the 1998 bleaching event. *Marine Ecology Progress Series* **212**, 301–304.
- Nakamura, T., Yamasaki, H. and van Woesik, R. (2003). Water flow facilitates recovery from bleaching in the coral *Stylophora pistillata*. *Marine Ecology Progress Series* **256**, 287–291.
- Navarrete, S. A., Gelcich, S. and Castilla, J. C. (2010). Long-term monitoring of coastal ecosystems at Las Cruces, Chile: Defining baselines to build ecological literacy in a world of change. *Revista Chilena de Historia Natural* **83**, 143–157.
- Norkko, J., Norkko, A., Thrush, S. F., Valanko, S. and Suurkuukka, H. (2010). Conditional responses to increasing scales of disturbance, and potential implications for threshold dynamics in soft-sediment communities. *Marine Ecology Progress Series* **413**, 253–266.
- Ojanguren, A. F., Reyes-Gavilán, F. G. and Braña, F. (2001). Thermal sensitivity of growth, food intake and activity of juvenile brown trout. *Journal of Thermal Biology* **26**, 165–170.
- Osman, R. W., Munguia, P. and Zajac, R. N. (2010). Ecological thresholds in marine communities: Theory, experiments and management. *Marine Ecology Progress Series* **413**, 185–187.
- Paine, R. T. (1966). Food web complexity and species diversity. *The American Naturalist* **100**, 368–378.
- Paine, R. T. (1969). The *Pisaster–Tegula* interaction: Prey patches, predator food preference, and intertidal community structure. *Ecology* **50**, 950–961.
- Paine, R. T. (1974). Intertidal community structure: Experimental studies on the relationship between a dominant competitor and its principal predator. *Oecologia* **15**, 93–120.
- Parnesan, C. (2006). Ecological and evolutionary responses to recent climate change. *Annual Review of Ecology Evolution and Systematics* **37**, 637–669.
- Parnesan, C. and Yohe, G. (2003). A globally coherent fingerprint of climate change impacts across natural systems. *Nature* **421**, 37–42.
- Parnesan, C., Duarte, C., Poloczanska, E., Richardson, A. J. and Singer, M. C. (2011). Overstretching attribution. *Nature Climate Change*, 2–4.
- Patterson, M. R. (1992). A chemical engineering view of cnidarian symbioses. *American Zoologist* **32**, 566–582.
- Pearson, G. A., Lago-Leston, A. and Mota, C. (2009). Frayed at the edges: Selective pressure and adaptive response to abiotic stressors are mismatched in low diversity edge populations. *Journal of Ecology* **97**, 450–462.
- Petes, L. E., Mouchka, M. E., Milston-Clements, R. H., Momoda, T. S. and Menge, B. A. (2008). Effects of environmental stress on intertidal mussels and their sea star predators. *Oecologia* **156**, 671–680.
- Petratis, P., Methratta, E., Rhile, E., Vidargas, N. and Dudgeon, S. (2009). Experimental confirmation of multiple community states in a marine ecosystem. *Oecologia* **161**, 139–148.

- Pfister, C. A., Wootton, J. T. and Neufeld, C. J. (2007). Relative roles of coastal and oceanic processes in determining physical and chemical characteristics of an intensively sampled nearshore system. *Limnology and Oceanography* **52**, 1767–1775.
- Pincebourde, S., Sanford, E. and Helmuth, B. (2008). Body temperature during low tide alters the feeding performance of a top intertidal predator. *Limnology and Oceanography* **53**, 1562–1573.
- Pincebourde, S., Sanford, E. and Helmuth, B. (2009). An intertidal sea star adjusts thermal inertia to avoid extreme body temperatures. *The American Naturalist* **174**, 890–897.
- Pineda, J. and López, M. (2002). Temperature, stratification and barnacle larval settlement in two Californian sites. *Continental Shelf Research* **22**, 1183–1198.
- Place, S. P., O'Donnell, M. J. and Hofmann, G. E. (2008). Gene expression in the intertidal mussel *Mytilus californianus*: Physiological response to environmental factors on a biogeographic scale. *Marine Ecology Progress Series* **356**, 1–14.
- Poloczanska, E. S., Hawkins, S. J., Southward, A. J. and Burrows, M. T. (2008). Modeling the response of populations of competing species to climate change. *Ecology* **89**, 3138–3149.
- Porter, W. P. and Gates, D. M. (1969). Thermodynamic equilibria of animals with environment. *Ecological Monographs* **39**, 227–244.
- Pörtner, H.-O. (2010). Oxygen- and capacity-limitation of thermal tolerance: A matrix for integrating climate-related stressor effects in marine ecosystems. *Journal of Experimental Biology* **213**, 881–893.
- Pörtner, H. O. and Farrell, A. P. (2008). Physiology and climate change. *Nature* **322**, 690–692.
- Pörtner, H. O., Hardewig, I. and Peck, L. S. (1999). Mitochondrial function and critical temperature in the Antarctic bivalve, *Laternula elliptica*. *Comparative Biochemistry and Physiology – Part A, Molecular & Integrative Physiology* **124**, 179–189.
- Pörtner, H. O., Bennett, A. F., Bozinovic, F., Clarke, A., Lardies, M. A., Lucassen, M., Pelster, B., Schiemer, F. and Stillman, J. H. (2006). Trade-offs in thermal adaptation: The need for a molecular to ecological integration. *Physiological and Biochemical Zoology* **79**, 295–313.
- Power, M. E., Tilman, D., Estes, J. A., Menge, B. A., Bond, W. J., Mills, L. S., Daily, G., Castilla, J. C., Lubchenco, J. and Paine, R. T. (1996). Challenges in the quest for key-stones. *Bioscience* **46**, 609–620.
- Rockstrom, J., Steffen, W., Noone, K., Persson, A., Chapin, F. S., Lambin, E. F., Lenton, T. M., Scheffer, M., Folke, C. Schellnhuber, H. J., *et al.* (2009). A safe operating space for humanity. *Nature* **461**, 472–475.
- Sagarin, R. D. and Somero, G. N. (2006). Complex patterns of expression of heat-shock protein 70 across the southern biogeographical ranges of the intertidal mussel *Mytilus californianus* and snail *Nucella ostrina*. *Journal of Biogeography* **33**, 622–630.
- Sanford, E. (1999). Regulation of keystone predation by small changes in ocean temperature. *Science* **283**, 2095–2097.
- Sanford, E. (2002a). Water temperature, predation, and the neglected role of physiological rate effects in rocky intertidal communities. *Integrative and Comparative Biology* **42**, 881–891.
- Sanford, E. (2002b). The feeding, growth, and energetics of two rocky intertidal predators (*Pisaster ochraceus* and *Nucella canaliculata*) under water temperatures simulating episodic upwelling. *Journal of Experimental Marine Biology and Ecology* **273**, 199–218.
- Sará, G., Kearney, M. and Helmuth, B. (2011). Combining heat-transfer and energy budget models to predict thermal stress in Mediterranean intertidal mussels. *Chemistry and Ecology* **27**, 135–145.
- Scheffer, M., Carpenter, S., Foley, J. A., Folke, C. and Walker, B. (2001). Catastrophic shifts in ecosystems. *Nature* **413**, 591–596.

- Schneider, K. R., Van Thiel, L. E. and Helmuth, B. (2010). Interactive effects of food availability and aerial body temperature on the survival of two intertidal *Mytilus* species. *Journal of Thermal Biology* **35**, 161–166.
- Schoch, G. C., Menge, B. A., Allison, G., Kavanaugh, M., Thompson, S. A. and Wood, S. (2006). Fifteen degrees of separation: Latitudinal gradients of rocky intertidal biota along the California Current. *Limnology and Oceanography* **51**, 2564–2585.
- Sebens, K. P. (2002). Energetic constraints, size gradients, and size limits in benthic marine invertebrates. *Integrative and Comparative Biology* **42**, 853–861.
- Shashar, N., Kinane, S., Jokiel, P. L. and Patterson, M. R. (1996). Hydromechanical boundary layers over a coral reef. *Journal of Experimental Marine Biology and Ecology* **199**, 17–28.
- Sibly, R. M. and Atkinson, D. (1994). How rearing temperature affects optimal adult size in ectotherms. *Functional Ecology* **8**, 486–493.
- Somero, G. N. (2005). Linking biogeography to physiology: Evolutionary and acclimatory adjustments of thermal limits. *Frontiers in Zoology* **2**, 1.
- Somero, G. N. (2010). The physiology of climate change: How potentials for acclimatization and genetic adaptation will determine ‘winners’ and ‘losers’. *Journal of Experimental Biology* **213**, 912–920.
- Sommer, A. and Pörtner, H. O. (1999). Exposure of *Arenicola marina* to extreme temperatures: Adaptive flexibility of a boreal and a subpolar population. *Marine Ecology Progress Series* **181**, 215–226.
- Southward, A. J. (1955a). On the behaviour of barnacles: II. The influence of habitat and tide-level on cirral activity. *Journal of the Marine Biological Association of the UK* **34**, 423–433.
- Southward, A. J. (1955b). On the behaviour of barnacles: I. The relation of cirral and other activities to temperature. *Journal of the Marine Biological Association of the UK* **34**, 403–422.
- Southward, A. J. (1957). On the behaviour of barnacles: III. Further observations on the influence of temperature and age on cirral activity. *Journal of the Marine Biological Association of the UK* **36**, 323–334.
- Stenseth, N. C. (2007). Canadian hare–lynx dynamics and climate variation: Need for further interdisciplinary work on the interface between ecology and climate. *Climate Research* **34**, 91–92.
- Stenseth, N. C., Mysterud, A., Ottersen, G., Hurrell, J. W., Chan, K.-S. and Lima, M. (2002). Ecological effects of climate fluctuations. *Science* **297**, 1292–1296.
- Stenseth, N. C., Ottersen, G., Hurrell, J. W., Mysterud, A., Lima, M., Chan, K.-S., Yoccoz, N. G. and Adlandsvik, B. (2003). Studying climate effects on ecology through the use of climate indices: The North Atlantic Oscillation, El Niño Southern Oscillation and beyond. *Proceedings of the Royal Society of London Series B* **270**, 2087–2096.
- Stoner, A. W. and Ray-Culp, M. (2000). Evidence for Allee effects in an over-harvested marine gastropod: Density-dependent mating and egg production. *Marine Ecology Progress Series* **202**, 297–302.
- Strong, D. R. (1992). Are trophic cascades all wet? Differentiation and donor-control in speciose ecosystems. *Ecology* **73**, 747–754.
- Suchanek, T. H. (1992). Extreme biodiversity in the marine environment: Mussel bed communities of *Mytilus californianus*. *Northwest Environmental Journal* **8**, 150–152.
- Thomas, F. I. M. and Atkinson, M. J. (1997). Ammonium uptake by coral reefs: Effects of water velocity and surface roughness on mass transfer. *Limnology and Oceanography* **42**, 81–88.
- Tomanek, L. (2008). The importance of physiological limits in determining biogeographical range shifts due to global climate change: The heat shock response. *Physiological and Biochemical Zoology* **81**, 709–717.

- Thompson, M., Drolet, D. and Himmelman, J. (2005). Localization of infaunal prey by the sea star *Leptasterias polaris*. *Marine Biology* **146**, 887–894.
- Thorson, G. (1950). Reproductive and larval ecology of marine bottom invertebrates. *Biological Review* **25**, 1–45.
- US Environmental Protection Agency (EPA), (2008). *Effects of Climate Change for Aquatic Invasive Species and Implications for Management and Research*. National Center for Environmental Assessment, Washington, DC, pp. 337.
- Wethey, D. S. (1983). Geographic limits and local zonation: The barnacles *Semibalanus* (*Balanus*) and *Chthamalus* in New England. *Biological Bulletin* **165**, 330–341.
- Wethey, D. S. (1984). Sun and shade mediate competition in the barnacles *Chthamalus* and *Semibalanus*: A field experiment. *Biological Bulletin* **167**, 176–185.
- Wethey, D. S. and Woodin, S. A. (2008). Ecological hindcasting of biogeographic responses to climate change in the European intertidal zone. *Hydrobiologia* **606**, 139–151.
- Wiens, J. J. and Graham, C. H. (2005). Niche conservatism: Integrating evolution, ecology, and conservation biology. *Annual Review of Ecology Evolution and Systematics* **36**, 519–539.
- Woodin, S. A. (1981). Disturbance and community structure in a shallow water sand flat. *Ecology* **62**, 1052–1066.
- Wootton, J. T., Pfister, C. A. and Forester, J. D. (2008). Dynamic patterns and ecological impacts of declining ocean pH in a high-resolution multi-year dataset. *Proceedings of the National Academy of Sciences of the United States of America* **105**, 18848–18853.
- Yamane, L. and Gilman, S. E. (2009). Opposite responses by an intertidal predator to increasing aquatic and aerial temperatures. *Marine Ecology Progress Series* **393**, 27–36.
- Zacherl, D., Gaines, S. D. and Lonhart, S. I. (2003). The limits to biogeographical distributions: Insights from the northward range extension of the marine snail, *Kelletia kelletii* (Forbes, 1852). *Journal of Biogeography* **30**, 913–924.